Understanding the Spatio-Temporal Behavior of Sentinel-1 SAR Vegetation Indices Over the Brazilian Savanna

Jales de Freitas Bussinguer, Gustavo Macedo de Mello Baptista, Edson Eyji Sano, and Fabrício Assis Leal

***Abstract*— The behavior of five synthetic aperture radar (SAR) vegetation indices, namely, the dual-pol radar vegetation index (DpRVI), radar vegetation index (RVI) dual-pol, polarimetric RVI (PRVI), dual-pol SAR vegetation index (DPSVI), and modified DPSVI (DPSVIm) was investigated. The indices were derived from Sentinel-1 data between 2017 and 2021 for three vegetation classes (Forest, Savanna, and Grassland) in the Brasilia National Park (BNP), Brazil. Temporal profile analysis showed that all indices followed a seasonal pattern directly linked with the rainfall regime in the study site. DpRVI and RVI were related to the rate of radar signal variation over the vegetation as a function of seasonality. PRVI, DPSVI, and DPSVIm were directly related to the classes' biomass levels and its seasonal variations over time. Precipitation and seasonal vegetation structure changes were relevant drivers of the shift and spread of the indices' spatial distributions over time. A phenomenon called the index equalization effect was observed mainly on rainy dates, shifting and spreading the classes' distributions to the same value ranges and degrading its separability. From a populational perspective, all indices presented associative intersections between all vegetation classes, despite the influence of seasonality. From a local perspective, DpRVI and RVI had disjoint intersections; PRVI maintained the associative relationships; and DPSVI and DPSVIm provided almost a complete separation between Forest and Grassland in all seasons. In both perspectives, DPSVIm outperformed the other indices in separating the classes by presenting the less than 72.15% average overlaps between all populational distributions and the highest H0 rejection rates, above 97% over time.**

***Index Terms*— C-band, Cerrado, stochastic distances, time series.**

I. Introduction

THE Brazilian Cerrado, the second largest biome in South America, has lost about half of its original cover [1], [2]. This biome occupies approximately 2 million km^2 in the central part of Brazil's territory (about 24% of the country's total area) and is the most biologically diverse savanna in the world [3]. Cerrado hosts more than 12 000 plant species [4] and one-third of Brazilian diversity, with a high level of endemism [5]. Considered one of the 34 hotspots for biodiversity conservation [6], this biome is under pressure for agricultural expansion and intensification [1].

The Cerrado's land use and land cover (LULC) map of 2021, produced by the MapBiomas project [7], indicates that approximately 53% of the total Brazilian Cerrado is covered with native vegetation. It also indicates that approximately 46% of Cerrado is covered with different land use types, mainly agriculture, aquaculture, and mining. According to the Brazilian clear-cut deforestation monitoring program (PRODES) [8], there was an increasing trend in vegetation suppression between 2019 and 2022, where a total of 10 700 km^2 of the deforested area was detected.

The Brazilian strategies to monitor Cerrado primarily rely on satellite optical imageries, which is quite challenging since they rely on cloud-free scene conditions. The biome has a well-marked climatic seasonality with a six-month dry season and a six-month wet season [9]. Its characteristic vegetation is a composition of a mosaic of different proportions of evergreen riparian forests, deciduous and semi-deciduous forests, savanna woodlands, shrublands, and grasslands [9].

Synthetic aperture radar (SAR) systems are attractive for mapping and monitoring changes in the vegetation cover since they can acquire images regardless of cloud coverage and Earth's surface solar illumination conditions. Previous studies have investigated the potential of SAR data to map and monitor Cerrado vegetation [9], [10], [11], [12], [13], [14], [15]. Most studies relied on backscattering information to characterize different vegetation types within Cerrado in various regions.

The increasing availability of SAR data led by the European Space Agency (ESA) C-band Sentinel-1 mission is allowing researchers to develop dual-pol SAR vegetation indices, often focused on monitoring agricultural lands [16], [17], [18], [19], [20]. SAR vegetation indices take advantage of the polarimetric properties of the interaction between the radar signal and the vegetation canopies to enhance the contrast between vegetation and other distributed targets, simplifying the analysis of SAR imagery. To the best of our knowledge, studies investigating the use of SAR vegetation indices to map the Cerrado vegetation have not been addressed yet.

This study focuses on the following scientific question: how do the dual-pol SAR vegetation indices characterize the native vegetation of Cerrado? This question assumes that,

Manuscript received 22 March 2023; revised 11 August 2023 and 23 November 2023; accepted 8 December 2023. Date of publication 25 March 2024; date of current version 9 April 2024. This work was supported by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), Brazil. *(Corresponding author: Jales de Freitas Bussinguer.)*

Jales de Freitas Bussinguer and Gustavo Macedo de Mello Baptista are with the Institute of Geosciences, University of Brasilia, Brasilia 70910-900, Brazil (e-mail: jalesbussinguer@gmail.com).

Edson Eyji Sano is with Brazilian Agricultural Research Corporation (Embrapa Cerrados), Planaltina, Brasilia 73301-970, Brazil.

Fabrício Assis Leal is with the Multidisciplinary Center, Federal University of Acre, Cruzeiro do Sul 69980-000, Brazil.

Digital Object Identifier 10.1109/TGRS.2024.3381468

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

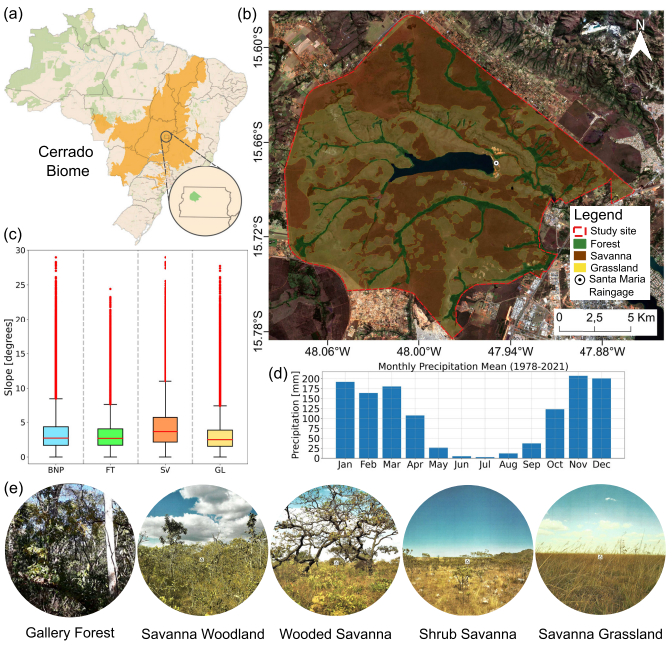


Fig. 1. (a) Location of the BNP in the Cerrado biome and in the Federal District of Brazil. (b) MapBiomas vegetation map of 2020 overlaid in a Sentinel-2 satellite image acquired on 06 July 2020 (RGB color composite of bands 4, 3, and 2). (c) Average monthly precipitation (1978–2021) data from the Santa Maria rain gauge available in the Hidroweb platform (www.snirh.gov.br/hidroweb/). (d) BNP and classes slope distributions based on SRTM 1 arc-second (30 meters) digital elevation model obtained from the Earth Explorer platform. (e) Field panoramic photos showing typical vegetation formations found in the park. Sources: [24] and [25].

by understanding the seasonal dynamics of Cerrado's native vegetation in terms of SAR data, it is possible to better identify and interpret potential human-induced or natural disturbances. Thus, the main purpose of this work is to investigate the potential of the dual-pol SAR vegetation indices to depict the seasonal dynamics of the Cerrado vegetation. Five Sentinel-1 SAR vegetation indices are explored to characterize the seasonality of three vegetation formations found in the Brasilia National Park (BNP), Brazil. The separability level between the three physiognomies in the study site was also assessed from populational and local perspectives to understand how seasonality influences the indices' spatial distributions over time.

II. Methodology

A. Study Site

The selected study site is the BNP, a 30 000-hectare preserved area, located in the north-western region of the Federal District of Brazil [see Fig. 1(a)]. The park is a relevant savanna research and conservation site, encompassing the major physiognomies of the Brazilian Cerrado [11]. The climate is characterized by a wet summer from October to April and a dry winter from May to September [10]. This seasonal pattern was considered in this study to evaluate its influence magnitude on SAR vegetation indices. The BNP topography is nearly flat to smoothly undulated [23]. About half the Shuttle Radar Topography Mission (SRTM) digital elevation model pixels in BNP's extent are below 8 degrees of slope [see Fig. 1(c)]. Hence, in this study, the topography influence in the SAR backscattering process was considered negligible.

Since the research area is permanently protected, no substantial land use changes have been found in this park over time. The BNP vegetation is representative of three major Cerrado physiognomies, referenced in this study as classes. Following the definition proposed by Ribeiro and Walter [24], the classes are Forest, Savanna, and Grassland, sorted by decreasing above-ground biomass content [22], respectively. The spatial extents of the classes were extracted from the MapBiomas Collection 6 LULC map of 2020.

The Forest class in the park represents the Gallery Forests physiognomy [see Fig. 1(b)], which occurs alongside brooks and streams as closed corridors of tall trees. They are characterized mainly by semi-deciduous to evergreen trees with 20–30 m average height with complex vegetation canopies [15], [24]. The predominantly evergreen condition of the Forest's vegetation maintains nearly constant high volumes of vegetation components like broad leaves and medium-sized branches in a dense and stable upper canopy. Due to the canopy's continuous dense biomass and tall trees, the interaction with the C-band signal is mostly limited to the upper part of the canopy. There is often a saturation of the C-band over the Forest class without underneath soil contribution to the total backscattered signal.

Savanna encompasses two physiognomies: Savanna Woodland and Wooded Savanna [10], [21], [24]. Savanna Woodland is composed of grasses, shrubs, and sparse trees mixture. The trees can reach an average height of 3–6 m with around 20%–50% of tree cover, forming a discontinuous canopy cover across the territory. The Wooded Savanna corresponds to a shrubland with sparse trees. The trees can reach an average height of around 2–3 m and around 5%–20% of tree cover. This vegetation unit represents the shortest and least dense class within the Savanna formation, with a predominance of shrubs and grasses.

The mixture of those two physiognomies within the Savanna formation results in a continuous shrub-herbaceous layer with a discontinuous tree layer composed of xeromorphic, semi-deciduous to evergreen trees clustered in different canopy densities [15], [22]. Some C-band signal saturation is expected in clustered areas of trees and where the composition of tree-shrub-grass density is high. Wherever the vegetation composition is less dense and sparse, some scattering contribution from the underneath soil is expected.

The Grassland class in BNP contains two Cerrado physiognomies, characterized by a predominance of grasses without woody species [10], [15], [24]. It is composed of Shrub Savanna and Savanna Grasslands. The Shrub Savanna is composed of the sparse presence of shrubs and herbaceous plants in open grasslands. The shrubby vegetation can reach an average height of 2 m and a tree cover of less than 5%. The grasses and herbaceous plants can reach an average height of 0.5 m. Savanna Grassland is an open grassland with the predominance of herbaceous plants and grasses, rare occurrences of shrubs, and a total absence of trees. Its vegetation cover can reach a mean height of 0.7 m, with grass stem elongations reaching 2.5 m [21].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

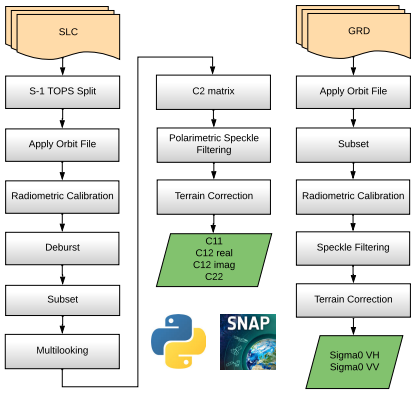


Fig. 2. Preprocessing steps for Sentinel-1 SLC and GRD products implemented in script using the SNAP API for Python.

The herbaceous plants present seasonal-marked phenological cycles. Its most vigorous phase occurs in the wet season, with the germination of new plants and the regrowth of the established ones by leaves and stems sprouting [25]. Because of its shallow roots, the water deficit in the dry season hinders the plant's growth and leads to its dissection, consequently forming a dry biomass [26]. A relatively low SAR scattering contribution is expected from plants and underlying soil due to low biomass density. In this case, its seasonal backscattering response may reflect contributions from the vegetation structure and its water content as well as from the soil roughness and moisture content.

B. SAR Data and Preprocessing Pipeline

The processing of the SAR vegetation indices relied on C-band Sentinel-1A products acquired in the interferometric wide swath (IW) mode in a descending orbit, with an average incidence angle of approximately 38.14°. The temporal dataset encompasses all single-look complex (SLC) and ground range detection (GRD) products available on the Alaska Satellite Facility (ASF) platform from January 2017 to December 2021. A total of 149 scenes with 12-day revisiting time were collected for each product. The Sentinel-1A products were used to calculate five SAR vegetation indices, namely, dual-pol radar vegetation index (DpRVI) [20], polarimetric radar vegetation index (PRVI) [16], radar vegetation index (RVI) [18], dual-pol SAR vegetation index (DPSVI) [17], and modified DPSVI (DPSVIm) [19]. The SLC products were used to calculate DpRVI and PRVI, while the GRD products were used to calculate RVI, DPSVI, and DPSVIm. Each pipeline's step depicted in Fig. 2 was executed in a fully automated batch mode, applied on each image separately by a Python script that implements the ESA's SNAP operators using its Python module Snappy. The link to access the source codes is available in Appendix A.

1) C_2 Matrix Components: At first, to obtain the C_2 matrix components, the identification of swaths and bursts of the SLC products that intersect the study site was made. The code parses the XML metadata of Sentinel-1.SAFE files and retrieves the geometries of scenes' swaths and bursts through the Python library S-1 TOPS Split Analyzer. The identified swaths and bursts were used as parameters for the TOPS Split operator. Then, the orbit state vectors were updated with precise orbit files. Subsequently, the products were radiometrically calibrated, and the output was saved in a complex-valued format. The calibrated products were then debursted and cropped using the study site spatial extent. The products were multilooked by 4×1 window size in range and azimuth directions to generate ground range square pixels. The produced C_2 matrix components were then speckle-filtered with the polarimetric Refined Lee filter [27] with a 3×3 kernel.

Finally, the C_2 matrix components were geocoded by the Range Doppler Terrain Correction operator into a Universal Transverse Mercator (UTM) projection system (EPSG:32722) and resampled to 10 m of pixel spacing with a bilinear interpolation algorithm. The final product was a Geotiff file with the C_2 matrix components (C_{11}, $\Re(C_{12})$, $\Im(C_{12})$, and C_{22}) expressed in power linear units.

2) Backscatter Coefficients: A preliminary step was comparing GRD products provided by ESA and GRD data derived from SLC products. The processing of GRD data from SLC products allows more control of the preprocessing parameters at the cost of more complex and extra steps than the GRD products provided by ESA. The comparisons of both GRD data showed slight differences between the backscattering coefficients' values on the decimal places. Those differences were assumed irrelevant to impact the results on the SAR vegetation indices implementation. Hence, the adoption of ESA's GRD products in this study was based on its straightforward preprocessing steps to obtain approximately the same result as those derived from SLC products.

The first step to compute σ^0_{VH} and σ^0_{VV} was to update the orbit state vectors of the GRD products. Then the imagery was clipped to the study site boundaries. The radiometric calibration was then applied to the VH and VV amplitude images. The speckle filtering was performed by a Refined Lee filter with a 3×3 kernel. Finally, the images were geocoded by the Range Doppler Terrain Correction operator. The final product was a Geotiff file with σ^0_{VV} and σ^0_{VH} data with 10 meters of pixel spacing, expressed in power linear units.

C. Temporal Profiles Analysis

First, the processing of all indices was made in a fully automated process implemented on a Python script. The script computes the indices from the SLC and GRD preprocessed products and stacks the results on the same raster file for each date. The source codes are available in Appendix A.

The pixel values of each class were extracted for each date to compose the probability density function (pdf) of the vegetation indices through time. Then, the extraction of zonal statistics such as median, standard deviation, upper quartile (q75), and lower quartile (q25) was made using the spatial extents of each vegetation class. These statistics were taken as a representation of the SAR vegetation indices' populational distribution variation over time. Due to the differences in scale between the Sentinel-1 data and the classes' spatial extents, a 20 m negative buffer was applied to each class's polygon to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

avoid the mixture of class interference in the zonal statistics. The Forest class encompassed 175 439 pixels, Savanna with 1 589 265 pixels, and Grassland with 1 105 350 pixels.

The term separability was employed in the temporal profiles analysis to represent the degree of distinction between vegetation classes with different biomass levels, which the indices can separate. Hence, to quantify the classes' separability for each vegetation index, the overlap coefficient (OVL) was computed.

Given two pdfs $f(x)$ and $g(x)$, the OVL is defined as

$$\mathrm{OVL}(f, g) = \int_{\mathbb{R}^n} \min(f(x), g(x))dx \tag{1}$$

which can be approximated for the discrete case as

$$\mathrm{OVL}(f, g) = \sum_{i=0}^{N} \min(f_i, g_i) \tag{2}$$

where f_i and g_i are the values of $f(x)$ and $g(x)$ known only at discrete positions x_i for $i = 0, \ldots, N$. The distributions were normalized so that $\sum_i f_i = \sum_i g_i = 1$. An OVL of 1 indicates the total superposition of the distributions, whereas an OVL of 0 means zero overlaps and therefore perfect separability.

A panel containing a time series of spatial median values with a surrounding polygon representing the interquartile ranges (IQRs), standard deviation, and OVL of all classes per date was constructed to observe and interpret the temporal variations of the SAR vegetation indices and their general trends. The OVL time series was plotted to observe the seasonality impact on the separability of the classes' populations. The indices maps and spatial distributions were also plotted in the panel to illustrate the seasonal differences in the vegetation indices' response. Daily precipitation data registered in the Santa Maria rain gauge, located in the study site, supported the analysis. The precipitation data was obtained from the Hidroweb platform.

D. Stochastic Distances Analysis

A series of hypothesis tests based on the Kullback–Leibler (KL) stochastic distances proposed by Nascimento et al. [28] was employed to understand how distinguishable the classes are over the seasons. Those experiments were conducted from a local perspective by using sample distributions of small pieces of the complete scene.

Stochastic distance is the distance between two probability distributions, and its concept is based on divergence measures. They are coefficients that reflect how close or far apart two distributions are from each other, with the property of becoming larger as the difference between distributions increases [29]. The majority of these coefficients lack the property of symmetry and so they are not truly distance metrics [29]. However, their symmetric variants are known as stochastic distances [28], which are suitable for comparing random variables since further distributions are easier to distinguish than closer ones. The KL stochastic distances can be interpreted as the separability between two distributions since smaller distances denote a higher degree of statistical similarities of the pdfs [30].

The assumption that different responses of the SAR indices regarding biomass level produce distinct spatial distributions throughout the seasons bases the use of this method to assess the local separability between classes. The null hypothesis (H0) of the test is that two sample distributions obtained in different regions of the study site can be described by the same distribution ($\widehat{\theta}_p = \widehat{\theta}_q$) [28].

To perform the series of hypothesis tests, first, a grid of 200 × 200 m^2 was constructed over the BNP vector in the QGIS software. From this grid, 50 squares of each vegetation class were randomly selected within the classes' spatial extent. The 200 × 200 m^2 were referenced in this study as sample patches of the complete scene. Every patch of every class contained 400 pixels (20 × 20 pixels) representing a local sample distribution of the vegetation indices.

The patches' size was constrained by Forest since it occurs along streams, and its maximum width is lower than 200 m in the terrain. The selected patches were used as masks in a Python script that reads the vegetation indices' rasters, calculates the sample distribution parameters, and saves the data in CSV files for each date.

In the literature, the stochastic distances assessment was employed for speckle noise reduction in SAR images by modeling the targets as complex Wishart distributions [31], Gamma distributions [32], or $\mathcal{G}^0$ distributions [28]. In this study, we assessed several distributions, and the log-normal distribution was the most suitable model for characterizing the vegetation indices' sample patch distributions for both seasons. Hence, to perform the hypothesis tests, the parameterization of the sample patch distributions was in terms of the mean (μ) and the standard deviation (σ) and thus represented by a parameter vector ($\widehat{\theta} = \mu, \sigma$).

The hypothesis tests were performed for each vegetation index separately, comparing the patch distributions of different pairs of classes in a step-wise approach implemented in a Python script. The script algorithm is given as follows.

1) Select the log-normal parameter vectors of two different patches from two different classes ($\widehat{\theta}_p = \mu_p, \sigma_p$ and $\widehat{\theta}_q = \mu_q, \sigma_q$).
2) Calculate the KL distance between parameter vectors (the proof is given in Appendix B)

$$d_{\mathrm{LN}}(\widehat{\theta}_p|\widehat{\theta}_q) = \frac{\sigma_p^2(\mu_p-\mu_q)^2+\sigma_q^2(\mu_q-\mu_p)^2+(\sigma_p^2-\sigma_q^2)^2}{4\sigma_p^2\sigma_q^2}. \tag{3}$$

3) Calculate the test statistic

$$s = \frac{2mnv}{m+n} d_{\mathrm{LN}}(\widehat{\theta}_p|\widehat{\theta}_q) \tag{4}$$

where m and n are the sample size of distributions p and q, respectively. v is a constant (1 for KL distances) [28].
4) Reject the null hypothesis if $Pr(\chi^2_M > s) \leq \eta$, where M is the number of degrees of freedom of the chi-square distribution, which is equal to the number of parameters of the target distributions. Since the distributions were parameterized in terms of mean and standard deviation,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

$M = 2$. η is the significance level of the tests, which was 0.05 in this study.

In the hypothesis tests, all possible pairs of patches of different classes were compared, totaling 7500 hypothesis tests per vegetation index and date. The null hypothesis rejection rates were calculated as the ratio between the H0 rejected tests and the total realized tests, which was adopted as a metric for the separability between the patch distributions. The rejection rates for each date were then plotted in temporal profiles for each vegetation index and pair of classes.

III. Results and Discussion

The potential of five C-band vegetation indices to describe the seasonal dynamics of three vegetation classes in the BNP is explored in this section. The results and discussion are organized as follows: temporal profile analysis of DpRVI, RVI, PRVI, DPSVI, and DPSVIm, respectively; comparison of the vegetation indices' populational separability based on the overlap coefficient; and comparison of the vegetation indices' local separability using a hypothesis test based on KL distances. In the following analyses, the term "biomass level" is related to the differences in the structural composition of these three vegetation formations, not directly related to their above-ground biomass quantity.

A. DpRVI Temporal Profiles

The DpRVI temporal profiles for each class are presented in Fig. 3(a). All classes showed a temporal development similar to bell-shaped curves following a seasonal pattern. The classes' medians increased in the wet seasons within an uptrend, typically reaching their highest values around October–November. The median values of DpRVI showed a downward trend during the dry season, typically reaching their lowest point around May–June. Within the seasonal pattern, the median curves were farther from each other in the wet seasons, returning to almost the same values by the end of the dry seasons. In the studied period, DpRVI presented typical values between 0.4 and 0.7.

The DpRVI profiles show an inversely proportional response to biomass level. Despite the similarly shaped curves, the three vegetation classes presented different behaviors regarding the median values. Forest presented almost constant median values, varying between 0.52 and 0.59, with an average value of 0.55 ± 0.017 (mean $\pm$ standard deviation). This class presented the lowest DpRVI values as well as the lowest variation amplitudes in the time series. Savanna also presented similar behavior but with more pronounced variation in the amplitude and an intermediate range of median values, predominantly concentrated between 0.53 and 0.63 with an average value of 0.57 ± 0.020. Grassland presented the highest median values and highest variation amplitudes in the time series, ranging between 0.54 and 0.66, with an average value of 0.59 ± 0.029.

The IQR polygons showed slight variations over time, an observation reinforced by the standard deviation profiles [see Fig. 3(b)]. The standard deviation presented slight fluctuations around 0.11 without a clear seasonal pattern. The

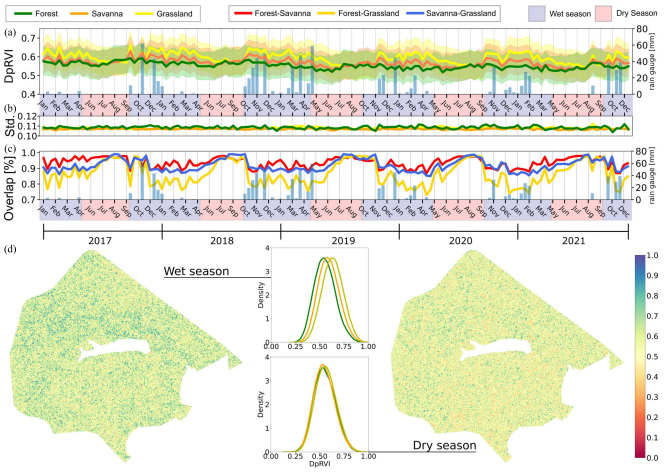


Fig. 3. (a) DpRVI temporal profiles of spatial median and IQR. (b) DpRVI standard deviation temporal profiles. (c) Temporal profiles of classes' overlap coefficients. (d) Study site DpRVI map from the wet season (22 December, 2020) and dry season (14 July, 2021). Precipitation is displayed as blue bars.

observed structural heterogeneity of the vegetation in different classes is probably responsible for the spatial variation observed in DpRVI values [20]. Hence, both IQR and standard deviation temporal behavior demonstrate that the seasonality influence only causes a shift in the classes' spatial distributions of DpRVI, not affecting its spread significantly over time.

The shifts in the classes' populational distributions can be observed by their overlap along the time series [see Fig. 3(c)]. The presence of seasonal variations can be seen in all pairs of classes, increasing in the dry seasons and decreasing in the wet seasons. The class distributions were further in the wet seasons, with a decrease in the overlap between the classes, especially on rainy dates. The pairs of classes were closer in the dry seasons, frequently reaching an overlap of almost 100%.

The DpRVI median profiles showed sensitivity to precipitation. There were more pronounced oscillations in the wet seasons' median values than in the dry seasons. On rainy dates, the precipitation increased the vegetation index values in all three classes, mainly observed as peaks in the median's profiles though the responses were distinct for each class. The lower the biomass level, the higher the amplitude of the peaks. The median values in Grassland presented the highest peaks, followed by Savanna and Forest, respectively. This observation might be respective to the classes' biomass levels and the influence of soil moisture in the total scattered signal. However, not all precipitation events increased the DpRVI values but maintained them at higher levels.

The difference in the peaks also had an impact on the separability of the classes. In Fig. 3(c), it is possible to observe that, on rainy dates, the overlap between the classes' distributions decreased but in different proportions. The shift in the median values caused by precipitation influence increased the separability between the classes' distributions. The larger shifts in Grassland medians turned it further from Forest and without significant variations over time.

It should be emphasized that there was a substantial overlap of over 70% among the categories across the entire time

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

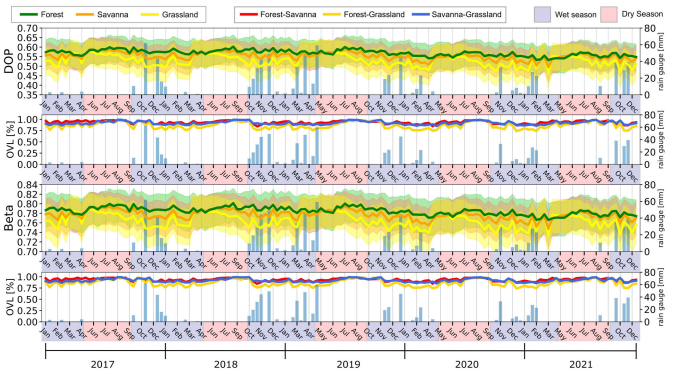


Fig. 4. DOP and β temporal profiles of spatial median and IQR. Precipitation is displayed as blue bars.

series. This scenario indicates that despite the influence of seasonality, the classes still carry a high level of overlap. The Forest–Grassland separability was the highest in the wet seasons, as the classes' distribution overlap was, in general, below 80%. The Forest–Savanna temporal separability presented similar behavior to that from Savanna–Grassland. Although there were minor variations in the exact overlap measurements during the wet seasons, both curves consistently shared overlaps of more than 85% throughout the time series.

The BNP presented DpRVI values of around 0.5–0.6 in both wet and dry seasons [see Fig. 3(d)]. The spatial distributions encompassing all pixels of the classes showed similar shapes. The main difference was the overlap between the classes' distributions. During the rainy season, the distributions of the classes were shifted from each other, though still with significant overlap. In the dry season example, all classes' distributions looked similar. Given the high overlap between classes' spatial distributions, the wet and dry season maps presented a similar fuzzy aspect, where the classes' spatial boundaries' visual recognition was not easy. Each class exhibited a consistent and cyclical seasonal pattern, with the main distinction being attributed to the impact of seasonal effects on vegetation structure (up/down trends) and dielectric properties (peaks).

The degree of polarization (DOP) and the scattering mechanism dominance (β) profiles (see Fig. 4) indicate that the three vegetation classes presented similar levels of canopy complexity along the time series, even considering seasonality effects. The DOP curves indicate that the biomass level is directly related to the medians and inversely related to the variations' amplitude. However, all classes cover almost the same range of values, concentrated between 0.38 and 0.66 along the time series. The similar overlap values between the three pairs of classes remained above 75% in the whole period, reinforcing the previous observations. The β curves presented the same temporal behavior, directly proportional to the biomass level, where the classes showed an overlap higher than 75% between their spatial distributions over time.

The DOP curves were directly proportional to biomass levels, demonstrating that more depolarization occurred with the decrease of the biomass level. With the increase in vegetation components and water content in the wet seasons, the variations in DpRVI values among the classes were apparent. However, all classes' median values remained below 0.6 over the seasons, indicating that the DOP only reached medium values even under seasonality effects.

Similar behavior was observed for the classes' β median curves, which remained above 0.7 over the seasons, indicating the predominance of a single dominant scattering mechanism. The classes seem to produce similar spatial distributions of DpRVI by presenting similar levels of DOP and scattering mechanisms dominance across seasons, leading to an overlap between the classes' distributions higher than 70%. A possible explanation for this scenario is that the surface roughness and canopy leaf layer contribute more to scattering at a higher mean incidence angle (such as 38.14° in this study), which reduces the effect of volume contribution in the backscattered signal [20].

The classes' temporal behavior of DOP and β might indicate the saturation of the C-band signal. The dense biomass of the Forest blocks C-band signal interactions with the underneath soil. Consequently, the signal saturation leads to similar levels of DOP in both seasons. The signal saturation effect is evident in the Forest curves of DOP and β values in Forest, with minimal seasonal variations that can be attributed to its evergreen condition. The saturation effect might be attenuated as biomass level decreases, leading to more pronounced variations of DOP and β values for Savanna and Grassland, likely due to the increasing influence of the underneath soil on the classes' backscattering process.

The visual difference between DpRVI maps calculated using ALOS/PALSAR-2 data (L-band) and Sentinel-1 (C-band) in [33] reinforces those assumptions. The L-band map displayed evident details and a broader range of values compared to the C-band map. Statistically, the L-band DpRVI outperformed the C-band DpRVI in a first-order microwave scattering algorithm to simulate the SAR backscattering response of the studied targets. Those observations point to a hypothesis that L-band DpRVI may effectively describe the different vegetation classes in BNP. Future SAR missions such as NISAR and BIOMASS would permit assessing this hypothesis.

In summary, as indicated by the profiles and maps, the seasonality indeed affects the classes' DpRVI response. Different responses regarding biomass levels and soil scattering influence were observed as the main drivers of the separability between classes, which can be enhanced in the wet seasons, especially on rainy dates. However, these differences are difficult to recognize, mainly based on visual interpretation of the maps. The stable temporal behavior and a minimum overlap of 70% along the time series make the classes difficult to separate.

Despite such stable seasonal behavior and high overlaps between the classes' populational distributions, DpRVI can be a suitable parameter for monitoring possible disturbances in the classes' vegetation structure. Monitoring strategies such as those described in [34] for the Brazilian Amazon may use DpRVI as a proxy for the temporal stability of the Cerrado physiognomies. In this context, any perturbation of the known temporal behavior within previously mapped classes'

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

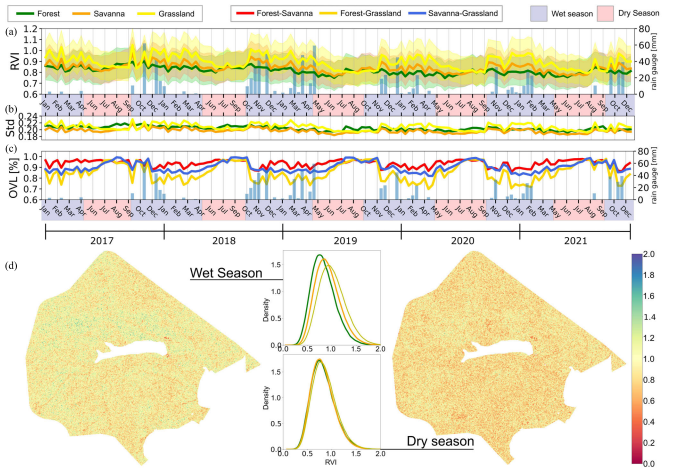


Fig. 5. (a) RVI temporal profiles of spatial median and IQR. (b) RVI standard deviation temporal profiles. (c) Temporal profiles of classes' overlap coefficients. (d) Study site RVI maps from the wet season (22 December, 2020) and dry season (14 July, 2021). Precipitation is displayed as blue bars.

extents can be considered an alert of disturbances like clear-cut deforestation in preserved Cerrado regions.

B. RVI Temporal Profiles

The RVI median profiles are shown in Fig. 5(a). The temporal profiles show a clear seasonal pattern, increasing in the wet seasons and decreasing in the dry seasons, presenting typical values between 0 and 2. The classes' median values increased within an uptrend in the first months of the wet seasons, reaching a plateau in the middle of the season. The index then presented a downward trend, typically reaching its lowest median values in the middle of the dry season, around July–August. The median curves were clearly separated from each other in the wet seasons. Toward the end of the dry seasons, they almost reached the same values, a similar behavior found for DpRVI.

The bell-shaped curves suggest an inversely proportional response of RVI mean in relation to the biomass level, where each class presented different variations over time. Forest presented an almost constant behavior as compared to the other classes, varying between 0.75 and 0.90 with an average of 0.81 ± 0.033. Savanna presented a profile similar to Forest, though with more pronounced variation in terms of amplitude, especially in the wet seasons. The Savanna medians were concentrated between 0.77 and 0.98 with an average of 0.85 ± 0.042, configuring an intermediate range of values. Grassland presented the highest median values and amplitude variation in the time series, especially in the wet seasons. Its medians were concentrated between 0.79 and 1.06, with an average of 0.90 ± 0.064.

The IQR values of the classes were similar throughout the time series, presenting a high level of overlap. The IQR polygons showed seasonal variations, which are reinforced by the classes' standard deviation curves [see Fig. 5(b)]. The standard deviation curves presented a temporal behavior similar to the medians with bell-shaped curves. Variation in amplitude was inversely proportional to the biomass level, suggesting that the index spatial variance decreases with the biomass increase. Both the IQR and standard deviation behaviors indicate that seasonality has an impact on the shift and spread of the RVI spatial distributions.

The impact of seasons on RVI can also be observed through the overlap of spatial distributions of different classes, which exhibited similar behavior to DpRVI. As shown in Fig. 5(c), the overlap between all pairs of classes' spatial distributions presented seasonal variations, increasing in the dry seasons and decreasing in the wet seasons. The distributions were further in the wet seasons, as the overlap between the classes decreased, especially on rainy dates. In the dry seasons, the pairs of classes were closer, frequently reaching an overlap of almost 100%.

The RVI median curves showed sensitivity to precipitation. In all profiles, the wet seasons showed more pronounced oscillations in the index median values than the dry seasons, where the curves were smoother. The wet seasons' oscillations were mainly linked to precipitation, observed as peaks in all profiles. However, not all precipitation events increased the index but maintained its higher levels, similar to the DpRVI curves. On rainy dates, precipitation led to an increase in the RVI values in all classes. However, the response was different regarding the biomass level and the influence of the soil in the total scattered signal. The lower the biomass level, the higher the amplitude of the peaks.

The difference in the rainy dates peaks also impacted the classes' separability [see Fig. 5(c)]. On the rainy dates, the overlap between the classes' distributions decreased, but in different proportions. The shift in the medians caused by precipitation influence increased the separability between the classes' distributions. The larger shifts in the Grassland medians turn it further from Forest, which do not have significant variations over time. It is possible to infer that the main driver of separability between the studied classes is the different rates of variation in the SAR signal, especially under the influence of precipitation.

The classes' overlap remained above 70% throughout the time series. Despite seasonal changes, there is still considerable overlap between classes, with a maximum separability of only 30%. The separability was the highest in the wet seasons, as the classes' distributions overlap was frequently below 90%. The Savanna–Grassland separability temporal development had similar behavior with the Forest–Grassland curve but with slightly higher overlap values. Although there were minor variations in the exact overlap values during the rainy season period, both curves maintained an overlap of over 70%.

The RVI values were typically around 0.7–1.2 for all classes in both seasons. The classes' spatial distributions showed similar shapes in both examples, where the main difference was the distribution overlap. In the wet season, the different class distributions appear to shift slightly, but there is still a significant overlap between them. In the dry season, all classes have very similar distributions. With the persistent high overlap between the classes' spatial distributions, both wet and dry seasons maps presented a similar fuzzy aspect, making it difficult to delineate the classes' borders [see Fig. 5(d)].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

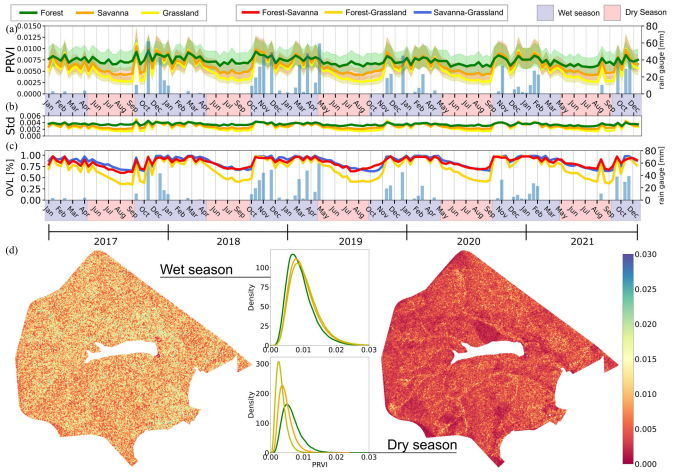


Fig. 6. (a) PRVI temporal profiles of spatial median and IQR. (b) PRVI standard deviation temporal profiles. (c) Temporal profiles of classes' overlap coefficients. (d) Study site PRVI maps from the wet season (22 December, 2020) and dry season (14 July, 2021). Precipitation is displayed as blue bars.

Since RVI is a measure of scattering randomness in the radar signal, its behavior is similar to DpRVI. Hence, RVI can also be considered a SAR parameter that expresses the variation rate of the radar signal over the classes. As the vegetation classes in BNP maintain high volumes of scatterers over time, the classes' scattering randomness may be mainly driven by seasonal influence in the vegetation phenologies. Since RVI is an unbounded index, the range of similarity between vegetations of different biomass levels reinforces the insight that RVI is not directly related to biomass level. The relationship is indirect by measuring the canopy complexity, often a consequence of biomass increase.

The findings on RVI temporal behavior indicate that seasonality indeed affects the index response. The different class response regarding its canopy complexity and soil scattering influence offers a low level of separability between the classes. During the rainy seasons, the distinction between classes was higher, particularly on dates with rainfall. However, the different classes' responses are hard to recognize, mainly based on visual interpretation.

The primary source of separability offered by RVI is the rate of variation in the radar signal for each class, particularly on days with rainfall. The different rates of variation can be a suitable parameter to measure the stability of the canopy complexity, which is an indirect measure of vegetation stability over time. In this context, disturbances in the baseline temporal behavior of preserved Cerrado regions caused by low-cut deforestation or forest fires can be considered an alert.

C. PRVI Temporal Profiles

Overall, the classes' PRVI profiles show a temporal behavior following a typical seasonal pattern with approximately bell-shaped curves over the seasons [see Fig. 6(a)], and typical values between 0 and 0.03. All PRVI medians increased in the wet seasons within an uptrend, reaching their highest values around October–November. During the dry season, the curves of the classes showed a consistent downward trend that typically ended between July and September. Throughout this period, the index consistently displayed its lowest values. Within the seasonal pattern, the medians' curves were farther from each other in the dry seasons, reaching almost the same values in the wet seasons, especially on rainy dates.

PRVI curves denote a directly proportional response regarding the biomass level. However, the amplitude of the classes' median fluctuations was higher with the lower biomass levels. Forest medians were almost constant compared to the other classes, varying between 0.0055 and 0.0090, with an average of 0.0068 $\pm$ 0.0008. Savanna medians presented a behavior similar to Forest, but with more pronounced fluctuation' amplitudes in an intermediate range of values, which were concentrated between 0.0038 and 0.0010 with an average of 0.0059 $\pm$ 0.0016. Grassland medians presented the lowest median values but the highest variation amplitudes over time, fluctuating between 0.0024 and 0.010 with an average of 0.0051 $\pm$ 0.0021.

The precipitation events affected the PRVI values over time. In the dry seasons, the PRVI values were directly proportional to the biomass level; i.e., the higher the biomass level, the higher the PRVI values. In contrast, wet season peak values reached similar values for all classes. The wet seasons showed more pronounced oscillations in the index medians than the dry seasons.

The PRVI peak was reached several times in the wet seasons, mostly on rainy dates, producing pronounced oscillations within the classes' bell-shaped curves. The amplitude of the peaks was inversely proportional to the biomass level, i.e., the lower the biomass level, the higher the amplitude of the spikes. The precipitation led to an abrupt increase in the index values to similar values in all profiles. The main difference between the classes was the magnitude of the peak.

The seasonal trends also evidence the influence of seasonality in the classes' behavior, which was likely related to vegetation structure changes. Within the wet seasons, the median values on dates without rainfall were the indicator of an uptrend, showing increases over time beyond the peaks on rainy dates. The increase in PRVI medians was possibly due to the increase in vegetation components, according to the vegetation phenology of each class, as a consequence of the higher soil moisture contents. The increase in the vegetation components probably makes the vegetation structures more random, enhancing their capacity to depolarize the radar signal.

The DOP curves in Fig. 4 show a decreasing trend within the wet seasons beyond the influence of the precipitation, reinforcing the previous observation. With the increase of the PRVI medians, the amplitude of the peaks decreased, especially for Grassland. These results suggest that vegetation structure constrains the precipitation influence on the PRVI values. On the other hand, in the dry seasons, the downtrends can be related to the decrease of vegetation components. Due to low levels of soil moisture content, the vegetation in Savanna and Grassland experiences some loss of components like leaves and stems, establishing a dry biomass [26].

The IQR was significantly different from each other along the time series. The IQR polygons around the median curves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

showed variations over time, suggesting that the distributions were more dispersed during the wet seasons compared to the dry seasons. These observations are reinforced by the classes' standard deviation curves shown in Fig. 6(b). The spread of the classes was directly proportional to the biomass level, suggesting that PRVI response in lower biomass levels is more homogeneous.

The spread of the classes followed a seasonal pattern, similar to the fluctuations in median curves. Such oscillations were inversely proportional to the biomass level, where Grassland showed the most pronounced seasonal variations, followed by Savanna and Forest, respectively. These observations indicate that rainfall seasonality affects both the shift and the spread of PRVI distributions over time, where wet seasons insert more uncertainty in PRVI than in the dry seasons.

The overlap between the populational spatial distributions evidences the influence of seasonality on PRVI. As shown in Fig. 6(c), the overlap between all pairs of classes' spatial distributions presented seasonal variations, increasing in the wet seasons and decreasing in the dry seasons. The pairs of classes were closer in the wet seasons, frequently reaching an overlap of almost 100%, especially by their end. The distributions of the classes were further in the dry seasons as their overlap decreased.

The classes' overlap presents different behavior along the time series. In the wet seasons, all three curves of overlap had a similar temporal profile with slightly different values. In contrast, the Forest–Grassland separability was the highest in the dry seasons, as their distribution overlap was frequently below 50%. There was a consistent temporal development in the Forest–Savanna and Savanna–Grassland curves, with both covering almost identical overlap values above 70%. This scenario indicates that, despite the influence of seasonality, both classes still carry a high level of overlap, with a maximum separability of 30%.

The spread of the classes' spatial distributions increased in the wet seasons, frequently reaching similar values in the rainy dates. It is possible to observe that the IQR values for all classes often covered almost the same range of values, with similar median values. In these dates, it is possible to observe peaks in the distributions overlap, often reaching 100%. These observations denote that the classes' distributions were almost equal in the spatial domain. The recurrence of these observations within the wet seasons suggests that the precipitation equalizes the PRVI response, shifting and spreading the distributions to approximately the same range of values. The shifts' amplitude was distinct for each class over time, but the median values did not surpass 0.10 along the time series. This context suggests that the precipitation raises the PRVI medians to a saturation point around 0.8–0.10. This phenomenon was called the index equalization effect in this study.

To illustrate the seasonal differences in the spatial domain, PRVI maps, and their respective classes' spatial distributions were depicted in Fig. 6(d). In the wet season, the classes' spatial distributions presented similar shapes covering approximately the same range of values. The high overlap between the classes' spatial distributions resulted in a map with a fuzzy

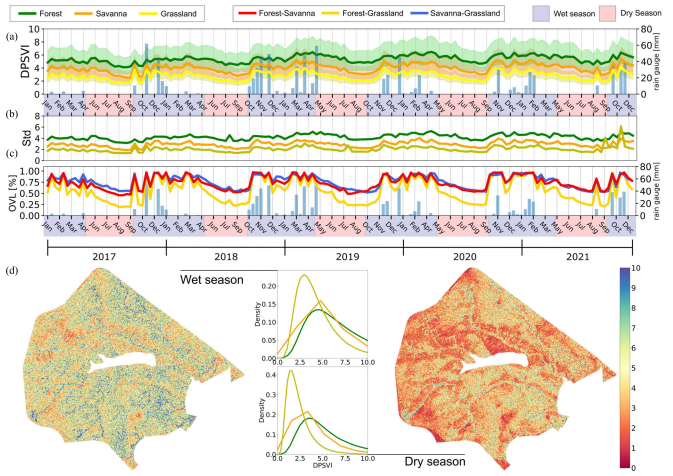


Fig. 7. (a) DPSVI temporal profiles of spatial median and IQR. (b) DPSVI standard deviation temporal profiles. (c) Temporal profiles of classes' overlap coefficients. (d) Study site DPSVI maps from wet season (22 December, 2020) and dry season (14 July, 2021). Precipitation is displayed as blue bars.

visual aspect without clear, recognizable spatial patterns. The index equalization effect degrades the classes' separability by shifting the classes' spatial distributions to almost equal levels. The dry season example presented significant differences between the classes' spatial distributions, with some overlap. The result was a map where the classes' spatial patterns are recognizable, with Forest represented in yellow, Savanna in orange, and Grassland in red.

The separability degree observed in the overlap profiles and the maps suggests that the class biomass level can be distinguished using PRVI. However, given the index temporal behavior, this separability is degraded in the wet seasons, given its sensitivity to precipitation. Its separability is enhanced in the dry seasons but maintains a high overlap between the classes' spatial distributions over time.

The PRVI temporal behavior showed a different behavior in comparison to DpRVI profiles, which also use DOP. This result might be possible due to a higher influence of σ^0_{VH} than the term (1-DOP) in the PRVI results. As analyzed before, the classes' DOP was directly proportional to the biomass level. In this case, the term 1-DOP inverts the curve trends, indicating that more depolarization of the radar signal occurs with the decrease in the biomass levels of the classes. By multiplying (1-DOP) by VH, the classes' curves assume the characteristic seasonal shape in BNP. It seems that σ_{VH} gives PRVI a sensitivity to biomass level and precipitation, which is probably the cause of its more pronounced fluctuations in the wet season. The oscillation of PRVI in the wet season might be due to the incorporation of σ_{VH} sensitiveness to the dielectric constant changes in the index formulation.

D. DPSVI Temporal Profiles

The spatio-temporal development of DPSVI profiles is shown in Fig. 7(a). All curves followed the seasonality trend patterns on the study site, where the index increased in the wet seasons and decreased in the dry seasons in definite bell-shaped curves. The wet seasons typically showed uptrends in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

all classes, with pronounced oscillations in the rainy dates. The dry seasons' temporal development occurred with lesser amplitude oscillations, typically reaching the lowest DPSVI median values. In the studied period, DPSVI presented typical values between 0 and 10, which contrasts with the 0–1 range reported by Periasamy [17].

Within the seasons' patterns, the classes' median curves were farther from each other in the dry seasons, reaching almost the same values in the wet seasons, especially on rainy dates. Luca et al. [35] investigated the postfire vegetation regrowth in Portugal and observed DPSVI interseasonal variation patterns mainly driven by humidity conditions, notably caused by precipitation over the study site.

All DPSVI median curves presented similar shapes, where the range of values was directly proportional to the biomass level. DPSVI values for Forest varied within the seasonal pattern with the highest medians, varying between 4.06 and 6.33 with an average of 5.30 ± 0.54 (mean $\pm$ standard deviation). Savanna varied within an intermediate range of medians, concentrated between 2.50 and 5.34 with an average of 3.96 ± 0.66. The Grassland class presented the lowest median values, varying between 1.70 and 4.13, with an average of 2.76 ± 0.61.

The amplitude of the variations increased with the decrease in the biomass level. Grassland presented the most pronounced variation amplitudes, followed by Savanna and Forest, respectively. These observations agree with the findings of [17], which found positive correlations between in situ AGB data and DPSVI in both seasons. The observed range of values assumed by the vegetation classes in BNP agrees with the statement that DPSVI values higher than 1 are representative of healthy vegetation [17]. The stratified range of values above 1 and directly proportional to the biomass level evidence the DPSVI capacity to map vegetation targets with different canopy complexities.

The IQR and the standard deviation curves [see Fig. 7(b)] suggest that the classes' distributions presented significant differences regarding the biomass level. Forest presented the highest IQR and standard deviation over the seasons, followed by Savanna and Grassland, respectively. This scenario suggests that the DPSVI of higher biomass levels carries more uncertainty than lower biomass levels in the spatial domain.

The IQR polygons around the medians presented slight variations, mainly following the shift of the medians. The standard deviation curves followed the seasonal pattern, but the amplitude of the variations was narrow. The almost constant temporal behavior of the classes' standard deviation curves evidences a slight difference in the distributions spread in wet and dry seasons. Those results suggest that seasonality predominantly affects the shift of the distributions, whereas the spread was not significantly affected over time.

The DPSVI median curves also indicate the sensitivity of the index to precipitation. During wet seasons, DPSVI typically reaches its peak on rainy dates. However, not all peaks reached the same median value on these dates. Despite this observation, the index equalization effect was observed in all rainy dates, where the classes' distributions overlap and often reach almost 100%.

The overlap between the DPSVI distributions also evidences the influence of seasonality on PRVI. As shown in Fig. 7(c), the overlap between all pairs of classes' spatial distributions presented seasonal variations, increasing in the wet seasons and decreasing in the dry seasons. During the wet seasons, the pairs of classes were often very similar, sometimes even overlapping almost completely by the end of the season. During the dry seasons, the distribution of the classes became more separated as their overlap decreased.

In the wet seasons, all three overlap profiles had a similar temporal development, with slightly different values often reaching almost 100%. During the dry seasons, the Forest–Grassland separability was the highest, as their distribution overlap was frequently below 30%. Forest–Savanna and Savanna–Grassland curves maintained similar temporal profiles, covering almost the same values of overlap above 50% throughout the time series. This scenario indicates that, despite the influence of seasonality, both classes still carry a high level of overlap, with a maximum separability of 50%.

To spatially depict the results, DPSVI maps and their respective classes' spatial distributions in both wet and dry seasons were produced [see Fig. 7(d)]. In the wet season example, it is possible to observe a high overlap between the spatial distribution of Forest and Savanna. Both classes showed higher values, displayed in violet-to-blue colors on the map. Their overlap resulted in a rough visual interpretation of these classes' patterns. The visual appearance of Forest and Savanna in the wet season map seems saturated in the range of 0–10. In the dry season example, the classes' distribution overlap decreased, enhancing their separability. The result was a map with an intelligible recognition of all classes' spatial patterns. Those observations agree with the maps produced by Periasamy [17], which differentiated sparse and dense vegetation into different classes.

In [17], the study focus was to assess DPSVI as a general vegetation descriptor, exploring its contrast with other targets such as water bodies and bare surfaces. The author found positive correlations between DPSVI and NDVI for both wet ($R^2 = 0.749$) and dry seasons ($R^2 = 0.726$), demonstrating the reliability of the index regarding seasonality. The results in the present study indicate that DPSVI has a well-suited capacity to distinguish vegetation covers with distinct biomass levels in BNP, despite the influence of seasonality. The medians remained separated most of the time, with the exception of some rainy dates.

Recent research validated the capacity of DPSVI for such analysis. Sosa et al. [36] used DPSVI and its derivatives (IDPDD, DPDD, and VDDPI) to analyze the impacts of hail damage in plots of corn, wheat, and soybean crops in Argentina. The analysis of 100 pixels within the study site presented a homogeneous DPSVI prestorm behavior. The results showed an increase in the variance after the storm, highlighting the index sensitivity to the backscatter changes caused by a hail storm in the studied plot. Luca et al. [37] investigated DPSVI as a vegetation descriptor to map burned areas in forested regions of Portugal and Italy. The index showed sensitivity to the backscatter changes after forest fires,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

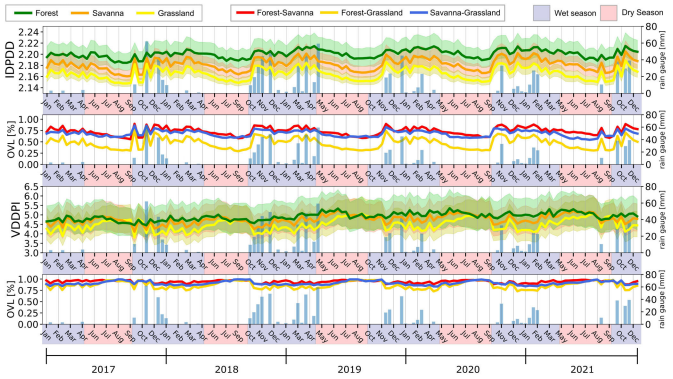


Fig. 8. IDPDD and VDDPI temporal profiles of the spatial median, IQR, and overlap coefficients. Precipitation is displayed as blue bars.

presenting well-defined areas of low backscatter in the burned regions.

To understand how DPSVI was capable of the observed separability levels between the vegetation classes, IDPDD and VDDPI temporal profiles were also constructed (see Fig. 8). IDPDD is a parameter that calculates Euclidean distances within the image scatter plot, based on the concept of the soil-vegetation edge [17]. Since the classes are representative of vegetations with different biomass levels, the curves were quite similar, with distinct degrees of overlapping between their spatial distributions over time.

During the dry seasons, Forest and Grassland were the most separated classes over time, often reaching a minimum overlap of 25%. The Forest–Savanna and Savanna–Grassland pairs had similar temporal development, presenting slightly different values of overlap. The degree of overlap between the different pairs of classes might be related to the limitation of IDPDD due to the saturation of the C-band over plants with more than a meter [17]. The IDPDD curves were directly proportional to the biomass level, similar to DPSVI. These observations indicate that IDPDD is an influential source of biomass-level separability within the DPSVI formulation.

The VDDPI and DpRVI curves show similar trends since the parameter estimates the rate at which the surface depolarizes the transmitted radar signal [17]. By using VDDPI, the targets that do not change the polarization of the illuminated signal occupy the lower range [17]. On the other hand, those that depolarize the signal occupy the higher range [17]. All three curves were highly overlapped over time (OVL $>$ 75%), denoting that the classes depolarize the radar signals similarly, as suggested by the DOP and β curves of Fig. 4. However, each curve had a different variation rate, evidencing the distinct influences of seasonality in the vegetation phenologies. These observations support that VDDPI is a parameter with sensitivity not to the biomass level but to the rate of variation in which the targets depolarize the radar signal.

The DPSVI temporal profiles showed that this index is well suited to express the seasonal patterns of the BNP vegetation. Different responses regarding the biomass levels of the classes and soil scattering contribution offer a separability suitable to map and model the above-ground biomass of Cerrado physiognomies. However, the index has an empirical characteristic by incorporating the parameter $\sigma^0_{VV_{max}}$ in IDPDD, which impairs its implementation in a time series analysis and on complete automated monitoring strategies of the Cerrado physiognomies.

Those limitations on DPSVI also impair the comparability between various regions since areas that differ in size and location may present different $\sigma^0_{VV_{max}}$ values. Moreover, the necessity of the scatter plots analysis inserts a source of human error in the model since the determination of $\sigma^0_{VV_{max}}$ is based on the judgment of the analysts. Despite its limitations, the analysis of DPSVI can be suitable to map disturbances such as clear-cut deforestation, forest fires, and hazard damages in preserved Cerrado regions.

E. DPSVIm Temporal Profiles

The DPSVIm median profiles showed a temporal development that followed the characteristic seasonal pattern in BNP [see Fig. 9(a)]. All classes' medians developed in approximately bell-shaped curves, increasing in the wet seasons within an uptrend, typically reaching their highest values around February–March. The index medians then began to decline in a downward trend which typically ended in the middle of the dry seasons, around August–September. Within the seasonal pattern, the median curves were farther from each other in the dry seasons, reaching almost the same values on rainy dates in the wet seasons. In the studied period, DPSVIm presented typical values between 0 and 0.1.

The DPSVIm temporal profiles also showed that the response of the index throughout the seasons was directly proportional to the biomass level of the classes. The Forest medians were the highest within the seasonal pattern, varying between 0.014 and 0.028, with an average of 0.021 $\pm$ 0.0035. Savanna occupied an intermediate range of values, where its median concentration over time was between 0.005 and 0.025, with an average of 0.013 $\pm$ 0.0050. The Grassland medians presented the lowest values of DPSVIm, but the highest variation amplitudes over time, fluctuating between 0.002 and 0.019, with an average of 0.008 $\pm$ 0.0044.

These results align with the findings reported by Santos [19], where DPSVIm had a directly proportional response regarding the biomass level of forested fragments in the Atlantic Forest biome in both seasons. Forest formations and planted forests had a higher value in their work, with annual crops and pastures closely behind. The index could also distinguish plots with biomass around 50.0 Mg ha^{-1} from those with around 200.0 Mg ha^{-1} in near-flat areas. These results are likely due to the enhanced sensitivity of DPDD to densely vegetated areas in comparison with IDPDD and the cross-ratio exponential behavior as a function of VDDPI in discriminating the increasing vegetation biomass levels [19].

The IQR was also directly proportional to the biomass levels, which indicates that the spatial heterogeneity decreases with the decreasing of biomass [see Fig. 9(a)]. The IQR polygons around the medians showed variations over time, indicating that wet seasons spread the distributions more than dry seasons. The classes' standard deviation curves in Fig. 9(b) reinforce those observations. The standard deviation curves showed an increase in wet seasons and a decrease in dry

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

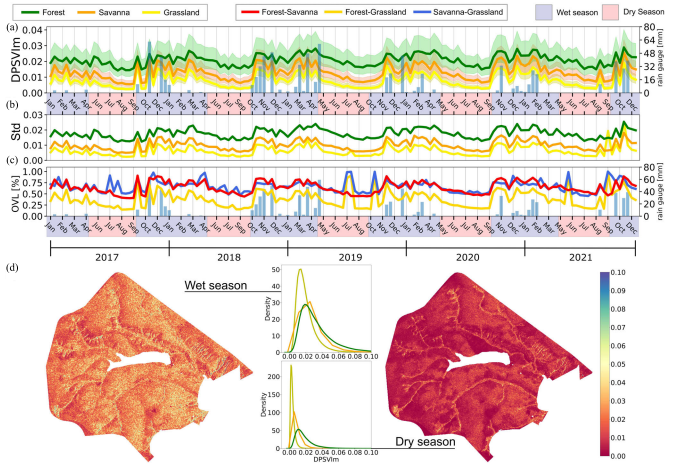


Fig. 9. (a) DPSVIm temporal profiles of spatial median and IQR. (b) DPSVIm standard deviation temporal profiles. (c) Temporal profiles of classes' overlap coefficients. (d) Study site DPSVIm maps from the wet season (22 December, 2020) and dry season (14 July, 2021).

seasons, with different variation amplitudes. The higher the biomass level, the lesser the variation amplitude. The variation amplitude difference between classes is likely due to the different levels of attenuation of the co-pol channel (VV) between classes' vegetations.

The seasonal behavior of the medians, IQR, and standard deviation indicate that rainy seasons introduced more variance in the distributions as a reflection of the increase of the surface moisture on the targets and components in the vegetation structure. The effects of seasonality not only shifted the DPSVIm distributions but also changed their spread over time. The rainy dates highlight these observations, where some precipitation events led to the sudden median and standard deviation peaks. According to [19], the DPSVIm also was affected by rainfall variations during the analyzed hydrological years, which caused uncertainties and resulted in lower correlations with spectral indices during the wet seasons.

The shifts and spreads in the DPSVIm spatial distributions can also be observed as their overlap over time [see Fig. 9(c)]. As for medians, the overlap between all class pairs showed seasonal variations, increasing in the wet seasons and decreasing in the dry seasons. The classes' distributions were further in the dry seasons, as their overlap decreased, often under 50%. On wet seasons, the pairs of classes were closer, frequently reaching an overlap of almost 100%, especially on rainy dates.

The DPSVIm median curves on the classes' temporal profiles indicate the index sensitivity to precipitation in BNP. On wet seasons, DPSVIm typically peaked on rainy dates of all analyzed years. However, not all peaks reached almost the same value on those dates. The medians reached almost the same value only after the first two days of rainfall in the 2017–2018 wet season. On other rainy dates, the distances between the classes' medians decreased but did not reach the same values. As observed in DPSVI results, the precipitation intensities along the time series seem not directly correlated with the index equalization effect in DPSVIm results.

The difference in the peaks also impacted the classes' separability [see Fig. 9(c)]. On rainy dates, the shift in the medians caused by precipitation influence decreased the separability between the classes' distributions but in different proportions. The larger shifts in Grassland medians turn it further from Forest, which had no significant variations over time. Therefore, these results suggest that the precipitation influence degrades the classes' separability.

Along the time series, except on some rainy dates in 2017, 2019, and 2021, the Forest–Grassland separability was the highest, as the distribution overlap was frequently below 25%. Forest–Savanna and Savanna–Grassland curves maintained a similar temporal profile, covering almost the same values of overlap, remaining above 50%. This scenario suggests that, despite the influence of seasonality, both classes still carry a high level of overlap and maximum separability of 50%.

It seems that DPSVIm attenuates the index equalization effect, maintaining higher levels of separability between the classes' spatial distributions despite the influence of seasonality. In [19], the increase in the spread of distributions in the wet seasons resulted in lower correlations of DPSVIm with NDVI and EVI. The authors also found that the agreement between DPSVIm and those spectral indices tends to increase by the end of the dry seasons, observed as peaks of positive correlations in the analyzed hydrological years. Those observations were probably a side effect of the index equalization.

A behavior often observed in all classes' profiles was the medians and standard deviation decrease in nonrainy dates after rainy dates. The amplitude decrease was distinct during the wet seasons, sometimes reaching DPSVIm values similar to those from dry seasons. These differences shorten during the wet seasons, evidencing an uptrend beyond the peaks caused by precipitation. An example is the 2017 dry-to-wet season transition. The uptrends presented a lag in the dry-to-wet season transition due to the time needed to increase the structural components in the vegetation, such as leaves, branches, and stems, after the first precipitations in the wet seasons. Hence, it is possible to infer that the structural changes in vegetation are responsible for the seasonal DPSVIm trends, whereas the precipitation is responsible for peaks in the DPSVIm values.

Those results highlight the capacity of DPSVIm to distinguish the vegetation cover by biomass level. The reflections of the seasonality interference in DPSVIm classes' distributions are also noticeable in the maps and the respective classes' distributions [see Fig. 9(d)]. In the wet season example, the classes' spatial distributions covered almost the same range of values. The overlap between the classes' spatial distributions resulted in a map with a fuzzy aspect but still with recognizable classes' spatial patterns, especially for Forest. Despite the index equalization effect on degrading the classes' separability, the wet season map evidenced the robustness of DPSVIm in maintaining some level of separability.

In the example of the dry season, there were noticeable variations in the spatial distributions of the different classes. The density curves were shifted from each other, also presenting different spreads, as observed before. The visual aspect of the map allowed a clear recognition of the classes' spatial patterns, which evidences the capacity of DPSVIm to distinguish the vegetation cover by biomass level.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

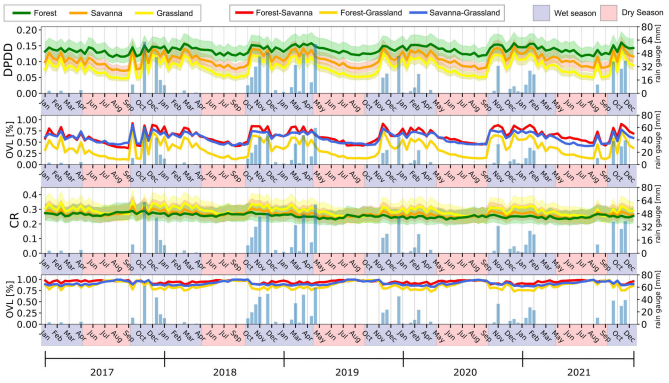


Fig. 10. DPDD and CR (VV/VH) temporal profiles of spatial median and IQR. Precipitation is displayed in blue bars.

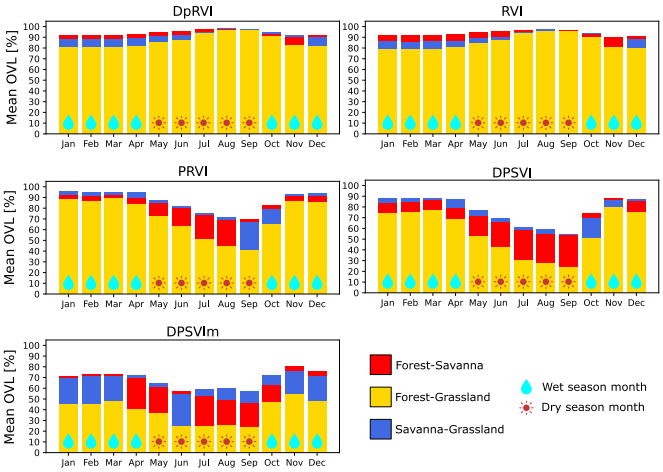


Fig. 11. Monthly average overlap coefficient between the classes' populational distributions. The decrease in the values indicates a higher separability.

The results of this study highlight the capacity of DPSVIm to distinguish vegetation covers with distinct biomass levels. The modifications proposed by Santos et al. [19] on DPSVI overcome its empirical characteristic, maintaining the separability between different biomass-level vegetations in both wet and dry seasons. Regarding these modifications, to understand how DPSVIm is capable of such apparent separability between classes, temporal profiles of DPDD and CR (VV/VH) were generated (see Fig. 10).

The DPDD curve shapes and IQR seasonal patterns were analogous to the variations of DPSVI and DPSVIm. The same is valid in terms of separability, where all three pairs of classes' curves had the same temporal behavior as the DPSVIm medians. These observations additionally indicate that DPDD has a strong influence on DPSVIm response. The DPDD curves also presented strong similarities with the IDPDD curves (see Fig. 8). However, the DPDD curves' variations presented a higher amplitude than IDPDD. This result demonstrates that DPDD presented an enhanced capacity to distinguish different biomass levels in densely vegetated areas compared to IDPDD, as discussed by Periasamy [17].

The spatial variability in DPSVIm response is likely due to the influence of DPDD as a measure of a diagonal Euclidean distance between the σ^0_{VV} and σ^0_{VH} response of the targets. In the Forest class, the VH backscatter is stronger than VV, and the diagonal distance is higher. Savanna presented a more balanced response in VV and VH due to higher soil influence. The variance is more likely to be controlled by the regions with higher density of trees. The variance in Grassland was the lowest due to approximately equal diagonal distances because of the contribution of the soil scattering in the total backscattered signal. The prevalence of grasses is evident in their stronger VV backscattering compared to VH.

The CR temporal profiles presented a temporal profile similar to DpRVI and RVI. The spatial distributions of CR showed a high overlap in the time series (OVL $>$ 0.75), where the main differences were the medians, following the same seasonal patterns as DpRVI, RVI, and the polarimetric variables DOP and β. Frison et al. [38] observed seasonal variations in CR over areas covered with pine and oak in France, attributing the variations mainly to seasonal variations in VH, whereas VV remained nearly constant over time. Based on current research, it seems that CR can be used to measure the rate a class's response changes because of seasonal factors. This principle is the same as that of DpRVI and RVI. The CR curves presented an inverse behavior with their respective VDDPI curves (see Fig. 8). It is possible to observe that CR presented a higher amplitude in its variations than VDDPI, despite showing similar trends with high overlap between its spatial distributions over time. Santos et al. [19] pointed out that CR has an exponential behavior as a function of VDDPI, with a higher potential to distinguish vegetation pixels.

DPSVIm temporal behavior showed that the seasonal patterns of the BNP vegetation are suitable for interpreting the index variations in a time series. From the classes' temporal profiles, it is possible to understand the temporal dynamics of the Brazilian Cerrado physiognomies and formulate strategies for monitoring the preserved areas. Given the potential good separability between the classes, the three major physiognomies of Brazilian Cerrado would be discriminable in supervised classification efforts. DPSVIm also can help reconstruct cloud-affected images from multispectral sensors, as investigated by Santos et al. [39]. The researchers found that DPSVIm played a significant role in reconstructing cloud-affected Landsat 8 imagery by modeling NDVI and EVI.

F. Vegetation Indices' Populational Separability Comparison

The temporal profiles show that the seasonality affects the classes' distribution shift (DpRVI, RVI, PRVI, DPSVI, and DPSVIm) and spread (PRVI, DPSVI, and DPSVIm). In this context, the overlap between the three pairs of classes changes with the seasons. The impact of seasonality on the indices' response predominantly influences its degree of separability. To show a detailed version of those observations from the temporal profiles and observe general trends of the indices, Fig. 11 depicts the monthly average overlap between the pairs of classes.

For DpRVI and RVI, the populational overlap increased in the dry seasons and decreased in the wet seasons in different proportions for each pair. During the wet seasons, the overlaps between the pairs of classes were maintained at higher levels

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

TABLE I
OVERLAP COEFFICIENT SEASONAL AVERAGES

OVL [%]	Forest-Savanna		Forest-Grassland		Savanna-Grassland	
	Wet	Dry	Wet	Dry	Wet	Dry
DpRVI	91.89	96.14	82.76	92.00	90.07	94.51
RVI	91.86	95.86	81.06	91.43	87.97	93.52
PRVI	89.75	74.65	83.23	54.05	91.87	76.14
DPSVI	82.74	60.31	71.36	35.04	84.78	64.16
DPSVIm	72.15	53.21	46.84	27.07	71.91	59.14

(OVL $>$ 80%) without significant variations. During the dry seasons, there is a noticeable increase in overlap between pairs of classes. The uptrend typically began in May and peaked in August and September. During these months, almost all pairs of classes had an overlap of around 99%.

PRVI, DPSVI, and DPSVIm distributions overlap increased in the wet seasons and decreased in the dry seasons (see Fig. 11). The results indicate that seasonality significantly impacts the overlap of the classes' populational distributions. The PRVI peak of separability was observed in September with an average overlap of around 40%. The DPSVI peak was observed in September with an average OVL of around 25%. The DPSVIm peak of separability occurred between June and September, with an average overlap of around 25%.

For these three indices, a downtrend can be observed starting in April, which is historically the least rainy month of the wet season. In the following months of the dry season, the average overlap continuously decreased, reaching its lowest average around August–September. With different proportions, those indices seem to be suitable parameters to observe the classes' differences over time, especially in the dry seasons. The wet seasons degrade the classes' separability, not being the best period to map the vegetation classes in BNP.

Table I provides the average overlap coefficient for each vegetation index and season, indicating the seasonal separabilities between the three pairs of classes. The table is color-coded on a linear scale, with green, yellow, and red corresponding to overlap coefficients of 0, 0.5, and 1, respectively.

As DpRVI and RVI quantify the SAR signal's rate of variation over the classes, both indices had similar seasonal average overlaps (see Table I). During the wet seasons, the Forest–Grassland pair had a minimum average overlap above 80%. The difference from the maximum average overlaps was around 10% for both indices. The average overlap for Forest–Savanna and Savanna–Grassland was above 90% in the wet and dry seasons. The difference from the maximum average overlaps was between 4% and 6%. Those results reinforce that both indices hold the least capacity to discriminate the classes despite the influence of seasonality, which makes the vegetation class classification a challenging task. The seasonality did not have a significant impact on the average DpRVI and RVI temporal behavior. Compared to their median temporal profile results, it seems that the spatial medians are the most suitable parameter to observe the three classes' differences over time.

For PRVI, each pair of classes had a different seasonal behavior (see Table I). The Forest–Savanna pair presented a minimum average overlap of 75% for the dry season, and the difference for the wet season was around 15%. The Forest–Grassland had the best separability average for the dry season (55%) and a difference to the wet season average of around 29%. The Savanna–Grassland presented a minimum average overlap of 76% and a difference from the wet season of almost 16%. Those results indicate that, on average, the separabilities of Forest and Grassland from Savanna are almost symmetrical in the dry seasons.

For DPSVI, each pair of classes also had a different seasonal behavior (see Table I). The Forest–Savanna pair presented a minimum average overlap of 60% for the dry season, and the difference for the wet season was around 12%. Forest–Grassland had the best separability average for the dry season, with 35% overlap, and a difference to the wet season average of around 36%. The Savanna–Grassland presented a minimum average overlap of 64% and a difference from the wet season of around 20%.

The best mean seasonal separability was presented by DPSVIm, as indicated in Table I. This index outperformed all the other indices by presenting the lowest values of average overlaps in both wet and dry seasons. The Forest–Savanna pair presented a minimum average overlap of 53% for the dry season, while the difference for the wet season was around 19%. The Forest–Grassland had the best separability average for the dry season, with 27% overlap, and a difference to the wet season average of almost 20%. The Savanna–Grassland presented a minimum average overlap of 59% and a difference from the wet season of almost 13%. This result is consistent with the results presented by the temporal profiles analysis and with the results reported by Santos et al. [19]. In this context, the DPSVIm can be considered the most suitable index to map the classes' population in the study site due to its enhanced capacity to discriminate the vegetation classes and the maintenance of this capability during almost all dry seasons.

The results of this section reinforce what was already apparent from the temporal profiles: on average, the vegetation indices have different capabilities to separate the studied vegetation classes. In addition, the indices showed that it was easier to distinguish between Forest and Grassland than between Forest and Savanna or between Savanna and Grassland.

The overlap averages results are also evidence of the kind of separability that each index can offer at a populational level. As the indices did not present a complete separability, it is possible to denote that all classes have associative intersections with each other (Forest $\cap$ Savanna, Savanna $\cap$ Grassland, and Grassland $\cap$ Forest) all over the time series. In this case, seasonality only influences the degree of intersection between classes.

G. Vegetation Indices' Local Separability Comparison

The temporal profiles of the five SAR vegetation indices presented a first approach to the separability between classes throughout the seasons. The indices' curves indicated that the seasonality effect shifts the classes' populational distributions to closer distances, mainly in wet seasons. PRVI, DPSVI, and DPSVIm presented an increase in class separability in the dry seasons, followed by a decrease in the wet seasons. Inversely,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

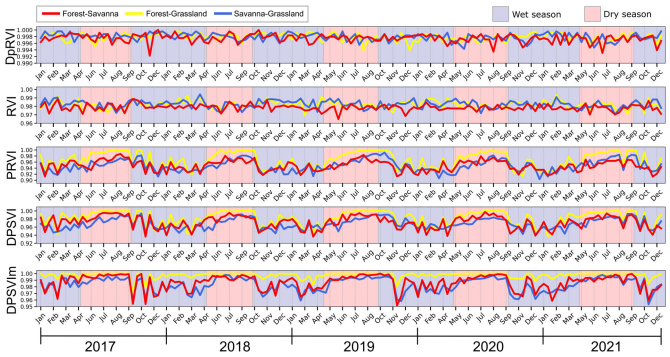


Fig. 12. Interclass profiles of H0 rejection rates ($p < 0.05$). The increase in the H0 rejection rate suggests an increase in the separability.

TABLE II
H0 REJECTION RATES SEASONAL AVERAGES

	Forest-Savanna		Forest-Grassland		Savanna-Grassland	
	Wet	Dry	Wet	Dry	Wet	Dry
DpRVI	99.81	99.75	99.81	99.75	99.81	99.75
RVI	97.89	97.82	98.35	98.02	98.37	98.07
PRVI	93.89	96.12	95.12	98.82	93.89	96.10
DPSVI	96.32	98.46	97.25	99.56	96.21	97.70
DPSVIm	98.10	99.47	99.63	99.99	97.73	99.16

DpRVI and RVI presented a seasonal behavior where the separability between classes increased in the wet seasons and decreased in the dry season.

The hypothesis tests based on stochastic distances were employed to assess how distinguishable the classes are from each other over the seasons from a local perspective. Fig. 12 shows the temporal profiles of H0 rejection rates between different pairs of class patches. Table II provides the average H0 rejection rates for each vegetation index and season. The table is color-coded on a linear scale, with green, yellow, and red corresponding to rates of 1, 0.5, and 0, respectively.

All curves of DpRVI indicate that the local separability between different pairs of classes is not significantly affected by the rainfall seasonality. The seasonal variations did not present any recognizable seasonal pattern and showed a slight difference between the temporal development for each pair. The lack of seasonal patterns in the H0 rejection rate curves might express the previously observed low sensitivity of the index to seasonal influences. The average rates of H0 rejection remained above 99.7% (see Table II) in all seasons for all pairs of classes, indicating that DpRVI patch distribution separability is almost complete over time.

The RVI curves indicate that the separability between the pairs of classes does not change significantly over time as a function of the rainfall seasonality, similar to DpRVI. The seasonal variations did not demonstrate a clear seasonal pattern, and the three pairs of classes had an almost constant temporal development. The index average rates of H0 rejection remained above 97.9% in all seasons for all pairs of classes (see Table II). This result also indicates that RVI sample distributions' separability is almost total over time.

The PRVI curves had a seasonality-influenced behavior where all pairs of classes' curves had similar temporal behavior, with a bell-shaped temporal development linked with the seasonality in the study site. The dry seasons showed higher rates of H0 rejection than the wet seasons, sometimes reaching an H0 rejection rate of 100%, especially for the Forest–Grassland pair. The decrease in H0 rejection rates in the wet seasons indicates that the seasonality effect approximated the PRVI patch distributions in the wet seasons. The index average rates of H0 rejection remained above 93.9% in the wet seasons and above 95.1% in the dry seasons. These were the lowest levels of H0 rejection between the studied indices.

The DPSVI curves presented a behavior where each pair had a different temporal development (see Fig. 12). The Forest–Savanna separability presented a clear seasonal pattern, decreasing in the wet seasons with an average of 96.3% of H0 rejection and increasing in the dry seasons with an average H0 rejection rate of 98.5%. The Forest–Grassland curve indicates an almost complete separability between the classes in the dry seasons, with an average H0 rejection rate of 97.3% in the wet seasons and 99.6% in the dry seasons. The pair Savanna–Grassland also developed within a seasonal pattern, increasing in the dry seasons with an average H0 rejection of 97.7% and decreasing in the wet seasons with an average H0 rejection of 96.2%.

For DPSVIm, similar to DPSVI, each pair of classes had a different behavior regarding seasonality, as shown in Fig. 12. The Forest–Savanna pair curve had a clear seasonal development, increasing in the dry seasons and decreasing in the wet seasons with an average H0 rejection of 98.1% in the wet seasons and 99.5% in the dry seasons. The Savanna–Grassland curve also shows a clear seasonal development, where the average H0 rejection rate was 99.8% throughout the seasons. The Forest–Grassland curve was nearly constant in the time series, where the H0 rates were almost 100%, with an average above 99% along the time series.

The pairs of curves for all indices presented high rates of H0 rejection over time, as the averages of H0 rejection were above 94% in both seasons. Those rates contrast with several studies that assessed the similarity between sample feature distributions and segments of SAR images [30], [40], [41]. For distinct model distributions and stochastic distances, those studies report H0 rejection rates below 1%. This divergence might be related to the distribution model selection. The log-normal curves might not be suitable to model the indices' distributions in the spatial domain. Further investigations are necessary to assess the model selection on the H0 rejection rates.

The H0 rejection rates' curves also evidence the kind of separability that the indices can offer from a local perspective, as indicated by Fig. 13. The DpRVI and RVI curves indicate a disjoint relationship of the three classes despite the seasonality effects (Forest $\cap$ Savanna $= \emptyset$, Savanna $\cap$ Grassland $= \emptyset$, and Grassland $\cap$ Forest $= \emptyset$). The PRVI curves denote that all classes have associative intersections with each other (Forest $\cap$ Savanna, Savanna $\cap$ Grassland, and Grassland $\cap$ Forest) all over the time series. DPSVI and DPSVIm provide a type of class separability where Forest and Grassland only intersect Savanna over time (Forest $\cap$ Savanna $\cap$ Grassland and Forest $\cap$ Grassland $= \emptyset$). For the last two cases, seasonality influences the degree of intersection between classes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

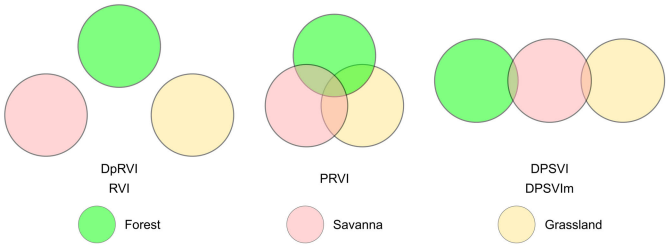


Fig. 13. Types of separability observed for the SAR vegetation indices sample distributions.

Those results point out a change in classes' separability from populational to local perspective for most of the indices. PRVI was the only index that preserved the associative intersections between all classes. DPSVI and DPSVIm progressed from associative intersections of all classes to a nearly complete separation of Forest and Grassland.

In specific DpRVI and RVI cases, there is a difference between the classes' separability observed from the overlap temporal profiles and the separability measured by the hypothesis tests. Their median, IQR, and overlap temporal profiles revealed a high degree of similarity between the populational distributions of the classes over time in associative intersections. However, both indices' H0 rejection rates show that sample distributions are almost completely separable across seasons. This scenario suggests that the observed H0 rejection rates are also influenced by the patch size of the classes (in pixels).

This assumption aligns with the results from [30]. The authors observed that the size of the samples influenced the rate of H0 rejection when comparing segments of L-band SIR-C simulated data against samples of nine different classes of actual SAR data. The results for KL distances showed that, as the segments' sizes increased, the percentage of simulated segments for which H0 was not rejected at a 5% significance level also increased. Qin et al. [42] found that the mean Kolmogorov–Smirnov distances between distributions modeled by Generalized Gamma, Gamma, and Weibull distributions decreased as the sample sizes increased. Silva and Nascimento [41] also observed that the H0 rejection rates decreased as the sample sizes increased for KL distances between $\mathcal{G}^0$ McKay bivariate distributions. If the same is valid for the case of this study, the increase in the patches' size might lead to a decrease in the H0 rejection rates.

Since the patches of 400 pixels used in this study only characterize local variations of those indices, the spatial locations of small pieces of the complete distributions can influence their distributions. The observed H0 rejection rates may be a result of the spatial variability of sample distributions. By increasing the size of the patches, the spatial variability influence on the patch distributions might decrease, leading to more acceptance of the H0. However, larger patches carry a higher mixture of random variables, which leads to undesired contamination of classes' patch distributions. The definition of patches with an equilibrium between statistical representativeness and the spatial context is a key issue in the reliability of the test results. Further investigations are necessary to understand the influence of the patch size on H0 rejection rates for the studied classes.

The speckle noise level in the imagery can be another influence on the high H0 rejection rates. It is possible that the level of speckle on C2 components and the backscattering coefficients contributes to adding randomness in the classes' index spatial distributions and thus influences test results. Further research is necessary to understand the indirect influence of speckle filters and their kernel sizes in the H0 rejection rates.

Despite its high values, all indices' H0 rejection curves validate the observations made from the temporal profiles and evidence potential applications for each index. The DpRVI and RVI's almost constant curves of H0 rejection rates indicate that both indices presented a low sensitivity to the seasonality effects on sample distributions, which did not significantly change the target's scattering processes. Those indices can be suitable to monitor defined spatial extents of Cerrado physiognomies as a proxy of the vegetation's temporal stability.

On the other hand, PRVI, DPSVI, and DPSVIm curves showed an explicit seasonal influence on the H0 rejection, which also validates the temporal profiles' observations. The spatial distributions of those indices were indeed affected by the rainfall regime and its influence on the phenological stages of the vegetation classes. The indices' sensitivity to different biomass levels grants an appropriate type of separability, which allows them to distinguish Forest and Grassland in both wet and dry seasons. Those types of separability have to be taken into consideration when selecting features for mapping the Cerrado physiognomies.

IV. Conclusion

This study explored the spatio-temporal behavior of five SAR vegetation indices over classes representing three Cerrado physiognomies: Forest, Savanna, and Grassland. The data was summarized in terms of spatial median, IQR, standard deviation, and the overlap coefficient between the classes' populational distributions that were extracted from every single date. The temporal profiles generated for the 2017–2021 period demonstrated the capacity of the indices to discriminate the vegetation in the BNP. For all vegetation index profiles, precipitation was found as the main driver for the peaks, while vegetation structure changes were the main driver for the shape of the curves.

DpRVI and RVI temporal followed the typical seasonal pattern of BNP, increasing in the wet seasons and decreasing in the dry seasons. However, the seasonality effect was insufficient to change their spatial distributions over time. The temporal profiles showed high overlaps between the classes' populational distributions of both indices over time, presenting low separabilities despite the seasonality influence. The maps came out with fuzzy aspects, without recognizable classes' spatial patterns in both wet and dry season examples. DpRVI and RVI can distinguish the vegetation classes by their rate of C-band signal variations, which is linked with their phenology and architecture.

The temporal behaviors of PRVI, DPSVIm, and DPSVI also followed the BNP seasonal pattern in approximately

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

bell-shaped curves, distinguishing the classes by the biomass level. In wet seasons, the increase in the indices values manifested as a shift in all classes' spatial distributions to almost the same levels, which led to lower separability between them. Especially on rainy dates, precipitation often raised all classes' medians to almost the same values, nearly to a saturation value. This phenomenon is called the index equalization effect, which degrades the classes' separability by increasing their distribution overlap. In contrast, the indices' values decreased, enhancing the classes' separability during the dry seasons. The example maps of all indices showed a more explicit differentiation between classes in the dry seasons than in the wet seasons, reinforcing this scenario.

The analysis based on KL stochastic distances validated the observations made with the temporal profiles. DpRVI and RVI H0 rejection rates' curves of all pairs of classes did not follow a clear seasonal pattern, evidencing that seasonality is not the principal driver for those indices' variations from a local perspective. PRVI, DPSVI, and DPSVIm H0 rejection rates followed the typical seasonal pattern in BNP, reinforcing that seasonality directly influences the indices' response in the spatial domain over time.

The indices were categorized into three groups. The first group is composed of DpRVI and RVI, presenting a completely disjoint relationship between the classes, contrasting with the observations made from the temporal profiles of overlap. Those indices' separability was hypothesized to be mainly influenced by the classes' size of sample distributions. The second group is composed only of PRVI, with associative intersections between classes all over the time series, both from populational and local perspectives. The third group contains DPSVI and DPSVIm, which showed a nearly complete separability between Forest and Grassland from a local perspective, while the degree of intersection was influenced by seasonality. Both classes only intersected Savanna along the DPSVIm time series, an evolution from the associative intersections of all classes in a populational perspective.

The main conclusion of this study is that the five SAR vegetation indices can effectively describe the Cerrado physiognomies. DpRVI and RVI have the utility potential as proxies of the vegetation temporal stability in previously mapped areas. PRVI, DPSVI, and DPSVIm have their utility potential as predictor variables to map the Cerrado physiognomies. The different potential of the indices to describe vegetation combined with the Sentinel-1' revisit time of 12 days configure a promising scenario to monitor the Cerrado physiognomies, especially in regions where the cloud cover decreases the availability of optical satellite data.

APPENDIX A
DATASETS AND SOURCE CODE LINKS

The data derived from the C2 matrix and backscatter coefficients imagery, as well as the Python scripts' source codes of preprocessing pipelines, vegetation indices processing, spatial statistics extraction, temporal profiles, and stochastic distance analysis notebooks, can be found in the following link: https://github.com/JalesBussinguer/sci_paper_code.

APPENDIX B
PROOF OF THE KL STOCHASTIC DISTANCE BETWEEN TWO LOG-NORMAL DISTRIBUTIONS

The KL divergence measure between two log-normal distributions, parameterized in terms of mean (μ) and standard deviation (σ) is [43]

$$D_{\mathrm{LN}}(\widehat{\theta}_p|\widehat{\theta}_q) = \ln\left(\frac{\sigma_p}{\sigma_q}\right) + \frac{1}{2\sigma_q^2}\left[(\mu_p-\mu_q)^2+\sigma_p^2-\sigma_q^2\right].$$

Since the stochastic distance is a symmetric version of the divergence measure [28]

$$d_{\mathrm{LN}}(\widehat{\theta}_p|\widehat{\theta}_q) = \frac{D(\widehat{\theta}_p|\widehat{\theta}_q)+D(\widehat{\theta}_q|\widehat{\theta}_p)}{2}.$$

The KL stochastic distance between two log-normal distributions is given by

$$\begin{aligned}
&d_{\mathrm{LN}}(\widehat{\theta}_p|\widehat{\theta}_q)\\
&= \left[\ln\left(\frac{\sigma_p}{\sigma_q}\right) + \frac{1}{2\sigma_q^2}\left[(\mu_p-\mu_q)^2+\sigma_p^2-\sigma_q^2\right]\right.\\
&\quad \left. + \ln\left(\frac{\sigma_q}{\sigma_p}\right) + \frac{1}{2\sigma_p^2}\left[(\mu_q-\mu_p)^2+\sigma_q^2-\sigma_p^2\right]\right] \times \frac{1}{2}\\
&= \left[\ln(\sigma_q) - \ln(\sigma_p) + \ln(\sigma_p) - \ln(\sigma_q)\right.\\
&\quad + \frac{1}{2\sigma_q^2\sigma_p^2}\times\sigma_p^2\left[(\mu_p-\mu_q)^2+\sigma_p^2-\sigma_q^2\right]\\
&\quad \left. + \sigma_q^2\left[(\mu_q-\mu_p)^2+\sigma_q^2-\sigma_p^2\right]\right] \times \frac{1}{2}\\
&= \frac{\sigma_p^2(\mu_p-\mu_q)^2+\sigma_q^2(\mu_q-\mu_q)^2+(\sigma_p^4-2\sigma_p^2\sigma_q^2+\sigma_q^4)}{4\sigma_q^2\sigma_p^2}.
\end{aligned}$$

Note that

$$a^2-2ab+b^2=(a-b)^2 \rightarrow (\sigma_p^4-2\sigma_p^2\sigma_q^2+\sigma_q^4)=(\sigma_p^2-\sigma_q^2)^2$$
$$a=\sigma_p^2, \quad b=\sigma_q^2.$$

Hence

$$d_{\mathrm{LN}}(\widehat{\theta}_p|\widehat{\theta}_q) = \frac{\sigma_p^2(\mu_p-\mu_q)^2+\sigma_q^2(\mu_q-\mu_p)^2+(\sigma_p^2-\sigma_q^2)^2}{4\sigma_p^2\sigma_q^2}.$$

ACKNOWLEDGMENT

The authors would like to thank the European Space Agency (ESA) and the Alaska Satellite Facility (ASF) for providing the Sentinel-1 SLC and GRD products through the ASF portal.

Special thanks to Andre Bindilatti for his contribution to the stochastic distances experimental design and code.

Another special thanks to Erli Santos for his insights and comments on the DPSVI and DPSVIm results.

This work is dedicated to the memory of Andre Matos.

REFERENCES

[1] E. E. Sano et al., "Land use dynamics in the Brazilian Cerrado in the period from 2002 to 2013," *Pesquisa Agropecuária Brasileira*, vol. 54, Apr. 2002, Art. no. e00138.

[2] B. B. N. Strassburg et al., "Moment of truth for the Cerrado hotspot," *Nature Ecol. Evol.*, vol. 1, no. 4, p. 99, Mar. 2017.

[3] M. M. C. Bustamante et al., "Potential impacts of climate change on biogeochemical functioning of Cerrado ecosystems," *Brazilian J. Biol.*, vol. 72, no. 3, pp. 655–671, 2012.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.

[4] C. A. Joly et al., *Apresentando o Diagnóstico Brasileiro De Biodiversidade e Serviços Ecossistêmicos*. São Carlos, Brazil: Editora Cubo, 2019.

[5] C. A. Klink and R. B. Machado, "Conservation of the Brazilian Cerrado," *Conservation Biol.*, vol. 19, no. 3, pp. 707–713, 2005.

[6] R. A. Mittermeier et al., *Hotspots Revisited: Earth's Biologically Richest and Most Endangered Terrestrial Ecoregions: Conservation International*. Sierra Madre, Cemex: Univ. Chicago Press, 2005.

[7] *MapBiomas Project—Brasil*. Accessed: Jun. 20, 2021. [Online]. Available: https://mapbiomas.org/en

[8] L. F. F. G. Assis et al., "TerraBrasilis: A spatial data analytics infrastructure for large-scale thematic mapping," *ISPRS Int. J. Geo-Inf.*, vol. 8, no. 11, p. 513, Nov. 2019.

[9] E. E. Sano et al., "Comparative analysis of the global forest/non-forest maps derived from SAR and optical sensors. Case studies from Brazilian Amazon and Cerrado biomes," *Remote Sens.*, vol. 13, no. 3, p. 367, Jan. 2021.

[10] E. E. Sano, G. G. C. Pinheiro, and P. R. Menezes, "Assessing JERS-1 synthetic aperture data for vegetation mapping in the Brazilian Savanna," *J. Remote Sens. Soc. Jpn.*, vol. 21, no. 2, pp. 158–167, 2001.

[11] E. E. Sano, L. G. Ferreira, and A. R. Huete, "Synthetic aperture radar (L band) and optical vegetation indices for discriminating the Brazilian savanna physiognomies: A comparative analysis," *Earth Interact.*, vol. 9, no. 15, pp. 1–15, Sep. 2005.

[12] F. D. S. Mendes, D. Baron, G. Gerold, V. Liesenberg, and S. Erasmi, "Optical and SAR remote sensing synergism for mapping vegetation types in the endangered Cerrado/Amazon ecotone of Nova Mutum—Mato Grosso," *Remote Sens.*, vol. 11, no. 10, p. 1161, May 2019.

[13] P. D. C. Bispo et al., "Woody aboveground biomass mapping of the Brazilian savanna with a multi-sensor and machine learning approach," *Remote Sens.*, vol. 12, no. 17, p. 2685, Aug. 2020.

[14] B. Zimbres et al., "Mapping the stock and spatial distribution of aboveground woody biomass in the native vegetation of the Brazilian Cerrado biome," *Forest Ecol. Manag.*, vol. 499, Nov. 2021, Art. no. 119615.

[15] K. Lewis et al., "Mapping native and non-native vegetation in the Brazilian Cerrado using freely available satellite products," *Sci. Rep.*, vol. 12, no. 1, p. 17, Jan. 2022.

[16] J. G. Chang, M. Shoshany, and Y. Oh, "Polarimetric radar vegetation index for biomass estimation in desert fringe ecosystems," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7102–7108, Dec. 2018.

[17] S. Periasamy, "Significance of dual polarimetric synthetic aperture radar in biomass retrieval: An attempt on Sentinel-1," *Remote Sens. Environ.*, vol. 217, pp. 537–549, Nov. 2018.

[18] R. Nasirzadehdijazi, F. B. Sanli, S. Abdikan, Z. Cakir, A. Sekertekin, and M. Ustuner, "Sensitivity analysis of multi-temporal Sentinel-1 SAR parameters to crop height and canopy coverage," *Appl. Sci.*, vol. 9, no. 4, p. 655, 2019.

[19] E. P. dos Santos, D. D. Da Silva, and C. H. do Amaral, "Vegetation cover monitoring in tropical regions using SAR-C dual-polarization index: Seasonal and spatial influences," *Int. J. Remote Sens.*, vol. 42, no. 19, pp. 7581–7609, Oct. 2021.

[20] D. Mandal et al., "Dual polarimetric radar vegetation index for crop growth monitoring using Sentinel-1 SAR data," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111954.

[21] R. D. Ottmar, R. E. Vihnanek, H. S. Miranda, M. N. Sato, and S. M. A. Andrade, "Stereo photo series for quantifying Cerrado fuels in central Brazil-volume I," U.S. Dept. Agricult., Forest Service, Pacific Northwest Res. Station, Portland, OR, USA, General Tech. Rep., PNW-GTR-519, 2001.

[22] M. B. T. da Costa et al., "Beyond trees: Mapping total aboveground biomass density in the Brazilian savanna using high-density UAV-LiDAR data," *Forest Ecol. Manag.*, vol. 491, Jul. 2021, Art. no. 119155.

[23] Serviço Nacional de Levantamento e Conservação de Solos, *Súmula da X Reunião Técnica De Levantamento De Solos*, Empresa Brasileira de Pesquisa Agropecuária (EMBRAPA), Rio de Janeiro, Brazil, Misc Series 1, Mar. 1979.

[24] J. F. Ribeiro and T. M. B. Walter, "The major physiognomy in the Brazilian Cerrado region," in *Cerrado: Ambiente e Flora*, S. M. Sano, S. P. Almeida, and J. F. Ribeiro, Eds. Planaltina, DF, Brazil: EMBRAPA-CPAC, 2008, ch. 3, pp. 169–188.

[25] G. Sarmiento and M. Monasterio, "Life forms and phenology," in *Ecosystems of the Worlds XIII*, F. Bourliere, Ed. Amsterdam, The Netherlands: Elsevier, 1983, ch. 5, pp. 79–108.

[26] P. E. A. M. Oliveira, "Phenology and reproductive biology of Cerrado's species," in *Cerrado: Ambiente e Flora*, S. Sano, S. Almeida, and J. F. Ribeiro, Eds. Brasília, DF, Brazil: EMBRAPA Cerrados, 2008, ch. 9, pp. 275–290.

[27] J.-S. Lee, "Refined filtering of image noise using local statistics," *Comput. Graph. Image Process.*, vol. 15, no. 4, pp. 380–389, Apr. 1981.

[28] A. D. C. Nascimento, R. J. Cintra, and A. C. Frery, "Hypothesis testing in speckled data with stochastic distances," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 1, pp. 373–385, Jan. 2010.

[29] A. A. Bindilatti and N. D. A. Mascarenhas, "A nonlocal Poisson denoising algorithm based on stochastic distances," *IEEE Signal Process. Lett.*, vol. 20, no. 11, pp. 1010–1013, Nov. 2013.

[30] W. B. Silva, C. C. Freitas, S. J. S. Sant'Anna, and A. C. Frery, "Classification of segments in PolSAR imagery by minimum stochastic distances between Wishart distributions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 3, pp. 1263–1273, Jun. 2013.

[31] N. Bouhlel, V. Akbari, and S. Méric, "Change detection in multilook polarimetric SAR imagery with determinant ratio test statistic," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5200515.

[32] J. Doblas, A. C. Frery, S. J. S. Sant'Anna, A. Carneiro, and Y. E. Shimabukuro, "Assessment of nonlocal means stochastic distances speckle reduction for SAR time series," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Feb. 2021, pp. 3265–3268.

[33] V. P. Yadav, R. Prasad, R. Bala, P. K. Srivastava, and V. S. K. Vanama, "Appraisal of dual polarimetric radar vegetation index in first order microwave scattering algorithm using Sentinel-1A (C-band) and ALOS-2 (L-band) SAR data," *Geocarto Int.*, vol. 37, no. 21, pp. 6232–6250, 2021.

[34] J. Doblas, Y. Shimabukuro, S. Sant'Anna, A. Carneiro, L. Aragao, and C. Almeida, "Optimizing near real-time detecion of deforestation on tropical rainforests using Sentinel-1 data," *Remote Sens.*, vol. 12, no. 23, p. 3922, 2020.

[35] G. Luca, J. M. N. Silva, and G. Modica, "Short-term temporal and spatial analysis for post-fire vegetation regrowth characterization and mapping in a Mediterranean ecosystem using optical and SAR image time-series," *Geocarto Int.*, vol. 37, no. 27, pp. 15428–15462, 2022.

[36] L. Sosa, A. Justel, and Ì. Molina, "Detection of crop hail damage with a machine learning algorithm using time series of remote sensing data," *Agronomy*, vol. 11, no. 10, p. 2078, Oct. 2021.

[37] G. De Luca, J. M. N. Silva, and G. Modica, "A workflow based on Sentinel-1 SAR data and open-source algorithms for unsupervised burned area detection in Mediterranean ecosystems," *GIScience Remote Sens.*, vol. 58, no. 4, pp. 516–541, May 2021.

[38] P.-L. Frison et al., "Potential of Sentinel-1 data for monitoring temperate mixed forest phenology," *Remote Sens.*, vol. 10, no. 12, p. 2049, 2018.

[39] E. P. dos Santos, D. D. da Silva, C. H. do Amaral, E. I. Fernandes-Filho, and R. L. S. Dias, "A machine learning approach to reconstruct cloudy affected vegetation indices imagery via data fusion from Sentinel-1 and Landsat 8," *Comput. Electron. Agricult.*, vol. 194, Mar. 2022, Art. no. 106753.

[40] A. Nascimento, J. Ferreira, and A. Silva, "Divergence-based tests for the bivariate gamma distribution applied to polarimetric synthetic aperture radar," *Stat. Papers*, vol. 64, no. 5, pp. 1439–1463, Oct. 2023.

[41] A. S. Silva and A. D. C. Nascimento, "A novel bivariate distribution and its divergence-based hypothesis inference: An application to the deforestation focus detection," *Int. J. Remote Sens.*, vol. 44, no. 1, pp. 1–29, Jan. 2023.

[42] X. Qin, H. Zou, S. Zhou, and K. Ji, "Divergence-based tests for the bivariate gamma distribution applied to polarimetric synthetic aperture radar," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 8, pp. 1655–1659, 2015.

[43] D. Dittrich, "Statistical methods for estimating loss distributions," M.S. thesis, Dept. Math., Technische Univ. München, München, Germany, 2013.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:15:49 UTC from IEEE Xplore. Restrictions apply.